

Proposal to Encode the Masaram Gondi Script in Unicode

Anshuman Pandey
Department of Linguistics
University of California, Berkeley
Berkeley, California, U.S.A.
anshuman.pandey@berkeley.edu

April 15, 2015

1 Introduction

This is a proposal to encode the Gondi script created by Mangal Singh Masaram in 1918 in Unicode. It replaces and supersedes the following documents:

- L2/10-207 “Preliminary Proposal to Encode the Gondi Script in the UCS”
- L2/12-235 “Revised Preliminary Proposal to Encode the Gondi Script”
- L2/15-005 “Proposal to Encode the Gondi Script”

This document provides a description of the writing system, a code chart and names list, character properties, and specimens that illustrate letterforms and usage. It is a revision of L2/15-005 and contains several changes to the encoding of Gondi proposed in that document. The major changes are as follows:

- The block name has been changed from ‘Gondi’ to ‘Masaram Gondi’
- Separate encoding of REPHA and RA-KARA for cluster-initial and cluster-final forms of RA
- Redefinition of VIRAMA specifically as a control character used only for producing conjuncts
- Addition of HALANTA as a combining sign used solely for silencing the inherent vowel
- Addition of CANDRA for transcribing foreign vowel sounds

These changes were introduced as a result of discussions with experts such as Mukund Gokhale, members of the user community and with Unicode engineers at Google and Microsoft.

The ‘Masaram Gondi’ script is graphically and structurally distinct from another Gondi script known as ‘Gunjala Gondi’. This latter script has been proposed for encoding in Unicode (see Pandey 2015b).

The symbol  appears upon several sources containing Masaram’s script and in various Gond religious contexts, in general. This symbol represents *persapen*, or the supreme spirit, in the indigenous Gond philosophical system known as *koya punem*. This symbol has been proposed for encoding in the ‘Miscellaneous Symbols and Pictographs’ block in Unicode, where several religious symbols are encoded (see Pandey 2015c).

2 Background

The script described here was invented by Munshi Mangal Singh Masaram of Kochewada, Balaghat District, Madhya Pradesh, India in 1918. It has no genetic relationship to other writing systems, but it is based upon the Brahmi model. The script was designed for writing Gondi (ISO 639-3: gon), a Dravidian language spoken by 2.6 million people, primarily in Madhya Pradesh and Maharashtra, with some speakers in Andhra Pradesh and Chhattisgarh. The language is generally written in both Devanagari and Telugu. Manuscripts containing yet another script, which appears to have been graphically inspired by these Modi writing system, was found in Gunjala in the Adilabad district of Telangana. Masaram's Gondi script is actively used today for hand-written and printed materials. Fonts have been developed for the production of books. In 2011, the Akhil Gondvana Gondi Sahitya Parishad (Chandagadh, Maharashtra) passed a resolution adopting Masaram's script as the official script of the Gondi language.

Masaram's script has been slightly expanded and revised over the years in order to meet the needs and preferences of modern users. Innovations include the addition of new consonant letters, vowel signs, modifier signs, and the adoption of a Devanagari-style *halanta* for indicating the absence of the inherent vowel. Some of these new characters are included in the proposed repertoire, while others are not (see section 4.11).

3 Script Details

3.1 Name

Earlier versions of this proposal referred to name of the script block as 'Gondi'. While the script is certainly used for writing Gondi, it is one of many scripts used for the language. Moreover, the 'Gunjala Gondi' script is also associated with the language and culture of the Gonds. Given this, it is appropriate to assign an identifier for the script block that precisely defines which 'Gondi' script is contained within that block. A designation that includes the name of the script's creator seems appropriate. For this reason, the name of the script block in Unicode is 'Masaram Gondi' and the names of characters reflect the block name. Users may refer to the script as 'Gondi' or by whatever name they prefer.

3.2 Structure

Masaram's Gondi script is an alphasyllabic writing system that is written from left to right. Consonant letters possess the inherent vowel *a*, which is graphically represented by a horizontal stroke that extends rightward from the right edge of each consonant letter. A bare consonant is represented by removing this stroke, particularly in the rendering of consonant clusters. Some modern users represent a word-final bare consonant by writing the *halanta* beneath the stroke of the consonant letter. Consonant clusters are represented as conjuncts, which are rendered as a linear sequence using bare forms for all consonants except for the final, which occurs in its regular form. There are some exceptions to this rule, namely the forms of *ra* when it occurs in initial and final positions in a cluster. Independent and initial vowels are written using vowel letters, while medial and final vowels are expressed using dependent signs. There is no *mātrā* reordering.

3.3 Character Repertoire

A total of 75 characters are proposed for encoding in the 'Masaram Gondi' script block. A code chart and names list are attached. Names for characters follow the UCS convention for Brahmi-based scripts and align with the Latin transliteration of Devanagari analogues for Gondi letters given by B. S. Masaram (1951).

3.4 Glyphic Representations

The glyphic representations of some consonant letters have changed since the invention of the script. The majority of changes are observed with consonant letters. These differences result from the simplification of glyphs for ease of writing, ie. sets of independent circles being joined into a single-stroked loop. Representative glyphs are based upon forms shown in published script primers and reflect modern preferences.

4 Proposed Encoding

4.1 Vowel Letters

Ten vowel letters are proposed for encoding:

𑌒	MASARAM GONDI LETTER A	𑌔	MASARAM GONDI LETTER UU
𑌓	MASARAM GONDI LETTER AA	𑌕	MASARAM GONDI LETTER E
𑌔	MASARAM GONDI LETTER I	𑌖	MASARAM GONDI LETTER AI
𑌕	MASARAM GONDI LETTER II	𑌗	MASARAM GONDI LETTER O
𑌖	MASARAM GONDI LETTER U	𑌘	MASARAM GONDI LETTER AU

Masaram's script does not have independent letters or dependent signs for the Dravidian long vowels /e:/ and /o:/, which correspond to Telugu ఌ ē and య ō.

4.2 Vowel Signs

Ten dependent vowel signs are proposed for encoding:

𑌐	MASARAM GONDI VOWEL SIGN AA	𑌑	MASARAM GONDI VOWEL SIGN VOCALIC R
𑌒	MASARAM GONDI VOWEL SIGN I	𑌓	MASARAM GONDI VOWEL SIGN E
𑌔	MASARAM GONDI VOWEL SIGN II	𑌕	MASARAM GONDI VOWEL SIGN AI
𑌖	MASARAM GONDI VOWEL SIGN U	𑌗	MASARAM GONDI VOWEL SIGN O
𑌘	MASARAM GONDI VOWEL SIGN UU	𑌙	MASARAM GONDI VOWEL SIGN AU

Vowel signs are written above and below the horizontal stroke of a consonant letter:

𑌒	𑌓	𑌔	𑌕	𑌖	𑌗	𑌘	𑌙	𑌚	𑌛	𑌜
ka	kā	ki	kī	ku	kū	kṛ	ke	kai	ko	kau

They are represented in encoded text as follows:

kā	𑌓	<𑌒 KA, 𑌐 SIGN AA>
ki	𑌔	<𑌒 KA, 𑌒 SIGN I>

<i>kī</i>	0 ^ᵃ	<0- KA, 0 ^ᵃ SIGN II>
<i>ku</i>	0 ^ᵁ	<0- KA, 0 ^ᵁ SIGN U>
<i>kū</i>	0 ^{ᵁᵁ}	<0- KA, 0 ^{ᵁᵁ} SIGN UU>
<i>kṛ</i>	0 ^{ᵁᵁ}	<0- KA, 0 ^{ᵁᵁ} SIGN VOCALIC R>
<i>ke</i>	0 ^{ᵁᵁ}	<0- KA, 0 ^{ᵁᵁ} SIGN E>
<i>kai</i>	0 ^{ᵁᵁᵁ}	<0- KA, 0 ^{ᵁᵁᵁ} SIGN AI>
<i>ko</i>	0 ^{ᵁᵁᵁ}	<0- KA, 0 ^{ᵁᵁᵁ} SIGN O>
<i>kau</i>	0 ^{ᵁᵁᵁᵁ}	<0- KA, 0 ^{ᵁᵁᵁᵁ} SIGN AU>

There is no independent analogue for VOWEL SIGN VOCALIC R. The independent form of this vowel is represented using a consonant-vowel combination composed with the letter RA:

<i>r</i>	0 ^{ᵁᵁᵁᵁ}	<0- RA, 0 ^{ᵁᵁᵁᵁ} VOWEL SIGN VOCALIC R>
----------	-------------------	---

4.3 Consonants

There are 34 consonant letters:

0-	MASARAM GONDI LETTER KA	ᵁ-	MASARAM GONDI LETTER DA
8-	MASARAM GONDI LETTER KHA	ᵁ-	MASARAM GONDI LETTER DHA
ᵁ-	MASARAM GONDI LETTER GA	ᵁ-	MASARAM GONDI LETTER NA
ᵁ-	MASARAM GONDI LETTER GHA	ᵁ-	MASARAM GONDI LETTER PA
ᵁ-	MASARAM GONDI LETTER NGGA	ᵁ-	MASARAM GONDI LETTER PHA
ᵁ-	MASARAM GONDI LETTER CA	ᵁ-	MASARAM GONDI LETTER BA
ᵁ-	MASARAM GONDI LETTER CHA	ᵁ-	MASARAM GONDI LETTER BHA
ᵁ-	MASARAM GONDI LETTER JA	ᵁ-	MASARAM GONDI LETTER MA
ᵁ-	MASARAM GONDI LETTER JHA	ᵁ-	MASARAM GONDI LETTER YA
ᵁ-	MASARAM GONDI LETTER NYA	ᵁ-	MASARAM GONDI LETTER RA
ᵁ-	MASARAM GONDI LETTER TTA	ᵁ-	MASARAM GONDI LETTER LA
ᵁ-	MASARAM GONDI LETTER TTHA	ᵁ-	MASARAM GONDI LETTER VA
ᵁ-	MASARAM GONDI LETTER DDA	ᵁ-	MASARAM GONDI LETTER SHA
ᵁ-	MASARAM GONDI LETTER DDHA	ᵁ-	MASARAM GONDI LETTER SSA
ᵁ-	MASARAM GONDI LETTER NNA	ᵁ-	MASARAM GONDI LETTER SA
ᵁ-	MASARAM GONDI LETTER TA	ᵁ-	MASARAM GONDI LETTER HA
ᵁ-	MASARAM GONDI LETTER THA	ᵁ-	MASARAM GONDI LETTER LLA

The letter 00- LLA is not part of Masaram's original script. It was introduced by modern users in order to properly represent Marathi ऌ *la* (see figure 19 for an example of LLA in usage).

kaḥ 𑌒 <𑌒 KA, 𑌐 VISARGA>

kāḥ 𑌒 <𑌒 KA, 𑌐 VOWEL SIGN AA, 𑌐 VISARGA>

Candra The sign 𑌐 CANDRA is used for transcribing vowel sounds that do not occur in Gondi. Examples are given in figure 25. It is derived from the sign 𑌐 used in Devanagari orthography for Marathi for representing the English vowel sounds /æ/ and /ɔ/. The CANDRA is written above the horizontal line of a consonant letter, and is used as follows:

/æ/ 𑌒 <𑌒 E, 𑌐 CANDRA>

/ɔ/ 𑌒 <𑌒 AA, 𑌐 CANDRA>

/kæ/ 𑌒 <𑌒 KA, 𑌐 CANDRA>

/kɔ/ 𑌒 <𑌒 KA, 𑌐 VOWEL SIGN AA, 𑌐 SIGN CANDRA>

In initial and independent contexts, CANDRA is used only with the vowel letters E and O for representing /æ/ and /ɔ/, respectively. The combination 𑌒 corresponds to 𑌒 U+090D DEVANAGARI LETTER CANDRA E, while 𑌒 corresponds to 𑌒 U+0911 DEVANAGARI LETTER CANDRA O. In dependent contexts, the CANDRA combines with a consonant letter for the sound /æ/, and with the VOWEL SIGN AA for the sound /ɔ/. The 𑌐 CANDRA corresponds to 𑌐 U+0945 DEVANAGARI VOWEL SIGN CANDRA E, while the sequence <𑌐 GONDI VOWEL SIGN AA, 𑌐 CANDRA> corresponds to 𑌐 U+0949 DEVANAGARI VOWEL SIGN CANDRA O. Although the Gondi sequences correspond to atomic characters in Devanagari, there is no need to encode such precomposed letters and signs with CANDRA for Gondi.

Although the contexts in which CANDRA is used are limited, the sign technically may be used with any letter and may occur with any vowel sign. It would be positioned with other vowel signs as follows:

𑌒 𑌒 𑌒 𑌒 𑌒 𑌒 𑌒 𑌒 𑌒 𑌒

4.5 Nukta

The 𑌐 MASARAM GONDI SIGN NUKTA is used for representing sounds that are not native to the Gondi language. It is written beneath the horizontal stroke of a consonant:

𑌒 𑌒 𑌒 𑌒 𑌒 𑌒 𑌒 𑌒 𑌒 𑌒

Some users prefer to position the NUKTA below the body of the consonant letter:

𑌒 𑌒 𑌒 𑌒 𑌒 𑌒 𑌒 𑌒 𑌒 𑌒

These positional preferences are to be managed in the font. The NUKTA is used in encoded text as follows:

ra ୨୮ <୨୮ DDA, ୦. NUKTA>
 rā ୨୮ <୨୮ DDA, ୦. NUKTA, ୦̣ SIGN AA>

4.6 Halanta

The ୦̣ MASARAM GONDI SIGN HALANTA is proposed as a vowel silencer. It is used as follows:

k ୦̣ <୦̣ KA, ୦̣ HALANTA>
 kh ୪̣ <୪̣ KHA, ୦̣ HALANTA>

The Gondi script as designed by Masaram does not have a native *halanta*, as the structure of the script does not require it. The embedding of the inherent vowel into the graphical structure of a consonant is a unique and innovative feature of the Gondi script. The horizontal stroke of each consonant letter represents the inherent vowel; removal of this stroke produces a bare consonant: ୦- *ka* → ୦ *k*, etc.

In most Indic scripts the inherent vowel is not part of the graphical structure of a consonant letter. As a result, these scripts require a mechanism for indicating the absence of the inherent vowel. In Devanagari this mechanism is a sign called ् U+094D DEVANAGARI SIGN VIRAMA: क *ka* + ् → क *k*, etc.

However, modern users have adopted the Devanagari *halanta* (or VIRAMA) for marking a bare consonant at the end of a word (see figure 24). The GONDI SIGN HALANTA has been included in the repertoire in order to provide this functionality in the proposed encoding.

The proposed encoding for Gondi separates the two functions of the VIRAMA character as used in the models for most Indic scripts in Unicode. In Devanagari, for example, the VIRAMA functions both as a vowel silencer (*halanta*) and a control character for forming conjuncts. The default representation of VIRAMA in Devanagari is as a visible sign beneath the consonant with which it combines. If a consonant is placed after the VIRAMA it causes a conjunct to be formed from the two consonants around it. In order to display a visible VIRAMA between adjacent consonants, it is necessary to break conjunct formation. For this purpose, the control character 𑀧 U+200C ZERO WIDTH NON-JOINER is placed after VIRAMA. This approach was proposed for Gondi in L2/15-005. However, in the interest of developing a simple encoding model for Gondi, and to eliminate the need for usage of ZERO WIDTH NON-JOINER or other invisible control characters, the encoding of the vowel silencing feature of the Indic VIRAMA has been encapsulated into the character ୦̣ GONDI SIGN HALANTA. The conjunct forming function is retained in the GONDI VIRAMA, described below.

4.7 Virama

The ୦̣ MASARAM GONDI VIRAMA is a control character that is used specifically for producing the bare form of a consonant letter. It is represented in the code chart as ୦̣ in order to indicate that it is a special character. Conceptually, VIRAMA produces a half-form by removing the horizontal stroke from the glyph of the letter after which it is placed.

k ୦ <୦ KA, ୦̣ VIRAMA>
 kh ୪ <୪ KHA, ୦̣ VIRAMA>

It is used for producing conjuncts, similar to the control function of ् U+094D DEVANAGARI SIGN VIRAMA. The Gondi VIRAMA, however, does not silence the inherent vowel; the ୦̣ GONDI SIGN HALANTA is to be used for that purpose.

4.8 Consonant Conjuncts

Consonant clusters are represented as conjuncts and are rendered in a linear sequence using half-forms of all but the final letter in a cluster, which appears in its full form, eg. 00- *kka*, 0U- *kta*, 0U8- *ktva*, etc. Consonants are placed sequentially in the conjunct in the order that they occur in the cluster.

Conjuncts are represented in encoded text by placing the control character  MASARAM GONDI VIRAMA after each non-initial consonant in a cluster. The sequence $\langle (C, \text{♀ VIRAMA})^*, C_f \rangle$ produces a half-form of all C and the regular full-form of C_f :

<i>ka</i>	0-	$\langle 0- KA \rangle$
<i>kta</i>	0U-	$\langle 0- KA, \text{♀ VIRAMA}, U- TA \rangle$
<i>ktva</i>	0U8-	$\langle 0- KA, \text{♀ VIRAMA}, U- TA, \text{♀ VIRAMA}, 8- VA \rangle$
<i>ktvya</i>	0U8-9-	$\langle 0- KA, \text{♀ VIRAMA}, U- TA, \text{♀ VIRAMA}, 8- VA, \text{♀ VIRAMA}, 9- YA \rangle$

There are some exceptions to the rule of conjunct formation. The following character are proposed in order to properly represent all Gondi conjuncts:

	MASARAM GONDI REPHA
○ ₁	MASARAM GONDI RA-KARA
𑌗-	MASARAM GONDI LETTER KSSA
𑌘-	MASARAM GONDI LETTER JNYA
𑌙-	MASARAM GONDI LETTER TRA

The behavior of RA is described in section 4.8.1, and the use of atomic ligatures for three conjuncts is discussed in see section 4.8.2).

4.8.1 Forms of RA in conjuncts

Following the general rule of conjunct formation, the letter 0- RA would occur as the half-form 0 when it is initial or medial in a cluster and in its regular full-form when in cluster-final position. The Gondi RA, however, does not behave entirely as expected in conjuncts and is rendered in several ways. There are three ways of writing 0- RA in conjuncts. It occurs in its half-form 0 when cluster-initial or alternately as 0[^] when cluster-initial and as ○₁ when cluster-final. These are described below:

- *Half-form* The half-form of RA is used specifically for representing semantic distinctions of RA when the letter occurs at a morphological boundary. Such usage is influenced by Devanagari orthography for the Marathi language, in which र RA may occur as either the ्र regular *repha* or the ्रँ ‘eyelash’ *repha* when it is the initial consonant in a cluster; the ‘eyelash’ *repha* marks plural suffixes (दर्या *daryā* ‘valleys’ and दर्या *daryā* ‘ocean’) and inflectional suffixes (आचार्यास *ācāryās* ‘to the cook’ and आचार्यास *ācāryās* ‘to the teacher’). The half-form 0 corresponds to the Devanagari ्रँ ‘eyelash’ *repha*. It is also used when RA occurs in cluster-medial position.
- *Repha* When not used for marking morphological distinctions, cluster-initial RA is rendered as the 0[^] *repha*. The logic of this character is based upon the Devanagari ्र regular *repha*. The Gondi *repha*

attaches after the last letter in a conjunct, above the horizontal line or an extension of it, depending upon the presence of an above-stroke vowel sign, eg. O^{a} *rka*, O^{a} *rkā*, etc.

O^{a} O^{a} O^{a} O^{a} O^{a} O^{a} O^{a} O^{a} O^{a} O^{a}

Some modern sources show the O^{a} *repha* represented using the form O^{a} , eg. O^{a} *rka*. This form is simply the regular sign for O^{a} *repha* with the left stroke drawn past the horizontal bar and curving to the right. It is a glyphic variant and is to be handled by the font.

- *Ra-kāra* In the current orthography, cluster-final RA is rendered as O_1 *ra-kāra* instead of as the full form O^- . The logic of the Gondi *ra-kāra* is based upon the Devanagari O *ra-kāra* and O *vattu*. The Gondi *ra-kāra* is positioned below the horizontal line of a consonant glyph or beneath an extension of the horizontal line:

O_1 O_1 O_1 O_1 O_1 O_1 O_1 O_1 O_1 O_1 O_1

Some modern sources show the O_1 *ra-kāra* represented using the form O_1 , eg. O_1 *kra*. This form is a glyphic variant and is to be handled by the font.

- *Full-form* The full-form of RA is rarely used at present when it is final in a cluster: OO^- *kra*. The preference is to use O_1 *ra-kāra*.

The representation of O^{a} *repha* and O_1 *ra-kāra* requires an exception to the rule of conjunct formation in Gondi. The general rule states that the sequence <RA, VIRAMA, C> is rendered using the half-form O of RA and the full-form of C. It also states that <C, VIRAMA, RA> would produce the half-form of C and the full-form O^- of RA. For this reason another method is required for the encoded representation of *repha* and *ra-kāra*, for which the expected encoded sequences according to the general model of Indic scripts would also be <RA, VIRAMA, C> and <C, VIRAMA, RA>, respectively.

There are four possible models for accommodating encoded representations of RA in conjuncts. The first two are based upon the premise that the default behavior of RA in conjuncts is similar to that of all other consonants, ie. it is rendered using the half-form O when cluster-initial and the full-form O^- when cluster-final. These approaches treat *repha* and *ra-kāra* as exceptions. The third model deviates from this premise and establishes the *repha* and *ra-kāra* as default representations of RA in conjuncts, and the half-form and full-form as exceptions.

1. Use the Zero-Width Joiner

In L2/15-005, it was suggested that the generic control character ZWJ U+200D ZERO WIDTH JOINER (ZWJ) be used for representing *repha* and *ra-kāra* in encoded text. The usage of ZWJ was chosen because the character is used in various Indic scripts for controlling different forms of letters in conjuncts. The same principle was applied to Gondi, such that various forms of RA in conjuncts would be produced as follows:

half-form	O	< O^- RA, O VIRAMA, C>
<i>repha</i>	O^{a}	< O^- RA, ZWJ ZERO WIDTH JOINER, O VIRAMA, C>
<i>ra-kāra</i>	O_1	<C, O VIRAMA, ZWJ ZERO WIDTH JOINER, O^- RA>

Representation of approach #1 in encoded sequences would be as follows:

$\textcircled{0}$ rka $\langle \textcircled{0} \text{ RA}, \text{♀ VIRAMA}, \textcircled{0} \text{ KA} \rangle$
 $\textcircled{0}$ kra $\langle \textcircled{0} \text{ KA}, \text{♀ VIRAMA}, \textcircled{0} \text{ RA} \rangle$
 $\textcircled{0}$ rka $\langle \textcircled{0} \text{ RA}, \text{ZWJ ZERO WIDTH JOINER}, \text{♀ VIRAMA}, \textcircled{0} \text{ KA} \rangle$
 $\textcircled{0}$ kra $\langle \textcircled{0} \text{ KA}, \text{♀ VIRAMA}, \text{ZWJ ZERO WIDTH JOINER}, \textcircled{0} \text{ RA} \rangle$

The Script Ad-hoc Committee recommended against the usage of ZWJ for such cases in Gondi. In L2/15-045, the Subcommittee suggested that an alternative model be developed and that “[f]or *repha* and *ra-kāra*, encoding separate characters may be useful, similar to the Malayalam *dot reph* and the *medial ra* in Myanmar and Tai Tham.” The major concern is that usage of ZWJ introduces issues for both implementers and end users. For implementers, the use of ZWJ overloads the script-specific rules that must be tailored for each script. For end-users, ZWJ is problematic because it is an invisible control character and is neither readily available on keyboards or easily detectable in text sequence.

2. Redefine the rule for rendering $\langle \text{RA}, \text{VIRAMA} \rangle$

The above model is based on the general rule of conjunct formation in Gondi: $\langle C, \text{VIRAMA} \rangle$ produces the half-form of C and therefore $\langle \text{RA}, \text{VIRAMA} \rangle$ should produce the half-form of RA . The *repha* and *ra-kāra* are treated as exceptions to the rule. A third approach for handling the different forms of RA in conjuncts is to redefine the default rendering for RA . With this approach the various forms of RA in conjuncts would be produced in a fashion similar to other Indic scripts:

$repha$ $\textcircled{0}$ $\langle \textcircled{0} \text{ RA}, \text{♀ VIRAMA}, C \rangle$
 $ra-kāra$ $\textcircled{0}$ $\langle C, \text{♀ VIRAMA}, \textcircled{0} \text{ RA} \rangle$

Representation of approach #2 in encoded sequences would be as follows:

$\textcircled{0}$ rka $\langle \textcircled{0} \text{ RA}, \text{♀ VIRAMA}, \textcircled{0} \text{ KA} \rangle$
 $\textcircled{0}$ kra $\langle \textcircled{0} \text{ KA}, \text{♀ VIRAMA}, \textcircled{0} \text{ RA} \rangle$

This approach, however, does not provide a means for producing the half-form $\textcircled{0}$ of RA , or even the cluster-final full-form if ever needed. Some mechanism would be required to produce these forms. Inevitably, a control character such as ZWJ would be required:

$\textcircled{0}$ rka $\langle \textcircled{0} \text{ RA}, \text{ZWJ ZERO WIDTH JOINER}, \text{♀ VIRAMA}, \textcircled{0} \text{ KA} \rangle$
 $\textcircled{0}$ kra $\langle \textcircled{0} \text{ KA}, \text{♀ VIRAMA}, \text{ZWJ ZERO WIDTH JOINER}, \textcircled{0} \text{ RA} \rangle$

This approach is essentially the converse of what was proposed in L2/15-005. It poses the same issues as it requires usage of ZWJ.

3. Encode a ligating form of RA

Another option is to encode a dummy letter whose cluster-initial form is $\textcircled{0}$ *repha* and whose non-initial form is $\textcircled{0}$ *ra-kāra*. This letter might be called LIGATING RA and it would be defined for usage

only in conjuncts. Another possible name might be RRA, but this name is generally used in Indic scripts in Unicode for the Dravidian *ra* and as Gondi is a Dravidian language, this name is reserved for potentially encoding a such a character for the Gondi script. The representative glyph for this letter might be , which is the letter 𑌒 RA placed within a dashed box that indicates that its function as a control character. The sequence <LIGATING RA, VIRAMA, C> would produce *repha*, while <C, VIRAMA, LIGATING RA> would produce *ra-kāra*. By extension, it does not have a half-form. The letter would be rendered as  when used in any other context. Using this character the various forms of RA would be produced as follows:

𑌒𑌒 *rka* <𑌒 RA, 𑌑 VIRAMA, 𑌒 KA>
 𑌒𑌒 *kra* <𑌒 KA, 𑌑 VIRAMA, 𑌒 RA>
 𑌒𑌐 *rka* < LIGATING RA, 𑌑 VIRAMA, 𑌒 KA>
 𑌒𑌐 *kra* <𑌒 KA, 𑌑 VIRAMA,  LIGATING RA>

Although LIGATING RA is an artificial letter and does not occur in the Gondi script, it facilitates the representation of *repha* and *ra-kāra* according to the general rules of conjunct formation, while also providing for the default rendering of RA as half-form and full-form in conjuncts:

𑌒𑌐𑌒 *rkra* < LIGATING RA, 𑌑 VIRAMA, 𑌒 KA, 𑌑 VIRAMA,  LIGATING RA>
 𑌒𑌒𑌒𑌒 *rkra* <𑌒 RA, 𑌑 VIRAMA, 𑌒 KA, 𑌑 VIRAMA, 𑌒 RA>

This approach also aligns with the concept that *repha* and *ra-kāra* in Devanagari and other scripts are special forms of a RA character. The approach that utilizes LIGATING RA offers users with a clear method for producing normative and special forms of RA in conjuncts using VIRAMA, which follows the general rule for the encoded representation of conjuncts in Gondi.

4. Encode ‘repha’ and ‘ra-kāra’ as separate characters

In L2/15-045, the Script Ad-hoc Committee recommended that 𑌒^h *repha* and 𑌒₁ *ra-kāra* be encoded as separate characters. While this approach provides the easiest means for encoding different forms of RA, it also raises several questions. Where should a *repha* character occur in an encoded sequence? Should it be handled logically and placed at the position where RA would normally occur, with the expectation that the rendering engine would reorder it to the end of the conjunct? Or should it be handled visually, and be placed manually at the end of the conjunct in an encoded sequence? For example, would the encoded representation for the hypothetical conjunct 𑌒𑌐 *rkra* be <𑌒^h *repha*, 𑌒 KA, 𑌒₁ *ra-kāra*> or <𑌒 KA, 𑌒₁ *ra-kāra*, 𑌒^h *repha*>? Another issue concerns the classification of *repha*: is it to be considered a letter (𑌒) or a sign (𑌒^h)? If *repha* is a letter, then is VIRAMA to be placed between it and the following consonant: <𑌒^h *repha*, 𑌑 VIRAMA, 𑌒 KA, 𑌒₁ *ra-kāra*>?

This approach requires the encoding of the following two characters:

 REPHA
 𑌒₁ RA-KARA

Furthermore, this approach requires the following considerations:

- (a) The *repha* is defined a letter, not a combining sign. In the code chart the [^] *repha* is placed within a dashed box as ^ in order to indicate that it requires special rendering. The glyph [^] is used in the output.
- (b) The logical model for *repha* is preferred over the visual model, ie. typing the character at the end of a cluster and any accompanying vowel signs. This approach provides the ability to type syllables according to the underlying phonology. The logical model requires that *repha* be placed at the same position in the encoded sequence that it occurs in phonetic expression. The rendering engine will re-order the *repha* to the end of the conjunct after any accompanying vowel signs.
- (c) The *ra-kāra* is a regular combining sign. It is to be placed, as expected, in its logical position before any accompanying vowel sign.
- (d) The VIRAMA is not to be used after either the *repha* or *ra-kāra*. Its presence in such contexts would be ignored, for example:

$\langle \text{^ REPHA, } \text{♀ VIRAMA, } \text{O- KA} \rangle \rightarrow \text{^ } \text{♀ O-}$
 $\langle \text{O- KA, } \text{♀ VIRAMA, } \text{○ RA-KARA} \rangle \rightarrow \text{O- } \text{♀ } \text{○}$

The REPHA and RA-KARA would be used in encoded sequences as follows:

$\text{O}^\text{a} \text{ rka} \quad \langle \text{^ REPHA, } \text{O- KA} \rangle$
 $\text{O}_\text{T} \text{ kra} \quad \langle \text{O- KA, } \text{○ RA-KARA} \rangle$

The usage of VIRAMA after and before RA would produce the expected output:

$\text{O O- rka} \quad \langle \text{O- RA, } \text{♀ VIRAMA, } \text{O- KA} \rangle$
 $\text{O O- kra} \quad \langle \text{O- KA, } \text{♀ VIRAMA, } \text{O- RA} \rangle$

The most feasible of the above four is approaches #4. Representing cluster-specific forms of RA as separate characters offers a simpler implementation that does not require usage of control characters. For this reason, the suggestion made by the Script Ad-hoc Committee in In L2/15-045 has been adopted.

4.8.2 Conjunct letters

The clusters *kṣa*, *jña*, *tra* are represented not as regular conjuncts, but as distinctive letters. These are proposed for encoding as atomic letters:

᳚ MASARAM GONDI LETTER KSSA
 ᳛ MASARAM GONDI LETTER JNYA
 ᳜ MASARAM GONDI LETTER TRA

Following the rules of conjunct formation, the expected representation of these three conjuncts would be:

kṣa 00- <0- KA, 0 VIRAMA, 0- SSA>
jña 01- <0- JA, 0 VIRAMA, 1- NYA>
tra U0- <U- TA, 0 VIRAMA, 0- RA>

In the Gondi script, each of these three letters phonetically represent a consonant cluster, but they all have the structure of an atomic letter. These forms are encoded as consonant letters because in all cases consonant conjuncts are written as linear sequences of half-forms, not as ligatures. While in most Indic scripts the written forms for *kṣa*, *jña*, *tra* have encoded representations as a character sequence, such an approach would not be consistent with this script.

It is evident that these ligatures were developed because distinctive forms exist in Devanagari. These three conjuncts are often shown at the end of Devanagari orthographies for various languages and are often interpreted by users as being distinctive letters that are fundamental elements of the script.

4.8.3 Rendering of conjuncts

As specified at the outset of this section, the sequence $\langle (C, 0 \text{ VIRAMA})^*, C_f \rangle$ produces a half-form of all C and the regular full-form of C_f . In order for the rendering of conjuncts to operate properly, the font must contain a complete set of consonants half-forms (glyphs without the horizontal stroke). The font should substitute each instance of $\langle (C, 0 \text{ VIRAMA}) \rangle$ with the appropriate half-form for C . If this glyph is not available, then the output should show the full form of C followed by 0 VIRAMA.

4.9 Digits

The script has a full set of digits:

0	MASARAM GONDI DIGIT ZERO	B	MASARAM GONDI DIGIT FIVE
1	MASARAM GONDI DIGIT ONE	6	MASARAM GONDI DIGIT SIX
2	MASARAM GONDI DIGIT TWO	J	MASARAM GONDI DIGIT SEVEN
3	MASARAM GONDI DIGIT THREE	9	MASARAM GONDI DIGIT EIGHT
4	MASARAM GONDI DIGIT FOUR	6	MASARAM GONDI DIGIT NINE

Variant forms of digits are shown in figure 18.

4.10 Punctuation

Script-specific punctuation is not attested in Masaram's Gondi script. The *daṇḍā* and double *daṇḍā* are commonly used. These signs of punctuation are not proposed for separate inclusion in the 'Masaram Gondi' block, but are to be unified with 1 U+0964 DEVANAGARI DANDA and 2 U+0965 DEVANAGARI DOUBLE DANDA. Latin marks of punctuation, such as periods, are also used.

4.11 Characters Not Proposed for Encoding

The following are newly-invented characters. Actual usage of these characters, apart from their inclusion in new charts of the script, is unknown. For this reason, they are not proposed for encoding at present.

Additional vocalic vowel signs The original script provided for the writing of only one Sanskrit vocalic vowel sign: ◌ᳵ VOWEL SIGN VOCALIC R. In order to accommodate anticipated usage of the script for linguistics research, Mukund Gokhale designed signs for the other three vocalic sounds:

- ◌ᳶ *GONDI VOWEL SIGN VOCALIC RR
- ◌᳷ *GONDI VOWEL SIGN VOCALIC L
- ◌᳸ *GONDI VOWEL SIGN VOCALIC LL

The sounds represented by these four vowel signs are not found natively in Gondi. They occur only in Sanskrit words, and only ◌ᳵ VOWEL SIGN VOCALIC R is attested in usage. There is no corresponding independent letter for these signs, and in such contexts they are represented using a consonant-vowel combination composed with the letters RA and LA and the dependent sign for the vocalic letter:

- ᳵ᳚ ◌ᳵ <᳚ RA, ◌ᳵ VOWEL SIGN VOCALIC R>
- *ᳶ᳚ ◌ᳶ <᳚ RA, ◌ᳶ *VOWEL SIGN VOCALIC RR>
- *᳷᳚ ◌᳷ <᳚ LA, ◌᳷ *VOWEL SIGN VOCALIC L>
- *᳸᳚ ◌᳸ <᳚ LA, ◌᳸ *VOWEL SIGN VOCALIC LL>

The VOWEL SIGN VOCALIC R is included in the proposed repertoire, but the other three signs are not proposed for inclusion at present because their actual usage is not attested. Space has been reserved for these characters in the block in the event that a justifiable case to encode them is made in the future.

5 Character Data

5.1 Character Properties

The properties for ‘Masaram Gondi’ in the Unicode Character Database format are:

```
11B90;MASARAM GONDI LETTER A;Lo;0;L;;;;N;;;;;
11B91;MASARAM GONDI LETTER AA;Lo;0;L;;;;N;;;;;
11B92;MASARAM GONDI LETTER I;Lo;0;L;;;;N;;;;;
11B93;MASARAM GONDI LETTER II;Lo;0;L;;;;N;;;;;
11B94;MASARAM GONDI LETTER U;Lo;0;L;;;;N;;;;;
11B95;MASARAM GONDI LETTER UU;Lo;0;L;;;;N;;;;;
11B96;MASARAM GONDI LETTER E;Lo;0;L;;;;N;;;;;
11B98;MASARAM GONDI LETTER AI;Lo;0;L;;;;N;;;;;
11B99;MASARAM GONDI LETTER O;Lo;0;L;;;;N;;;;;
11B9B;MASARAM GONDI LETTER AU;Lo;0;L;;;;N;;;;;
11B9C;MASARAM GONDI LETTER KA;Lo;0;L;;;;N;;;;;
11B9D;MASARAM GONDI LETTER KHA;Lo;0;L;;;;N;;;;;
11B9E;MASARAM GONDI LETTER GA;Lo;0;L;;;;N;;;;;
11B9F;MASARAM GONDI LETTER GHA;Lo;0;L;;;;N;;;;;
11BA0;MASARAM GONDI LETTER NGA;Lo;0;L;;;;N;;;;;
11BA1;MASARAM GONDI LETTER CA;Lo;0;L;;;;N;;;;;
11BA2;MASARAM GONDI LETTER CHA;Lo;0;L;;;;N;;;;;
11BA3;MASARAM GONDI LETTER JA;Lo;0;L;;;;N;;;;;
11BA4;MASARAM GONDI LETTER JHA;Lo;0;L;;;;N;;;;;
11BA5;MASARAM GONDI LETTER NYA;Lo;0;L;;;;N;;;;;
11BA6;MASARAM GONDI LETTER TTA;Lo;0;L;;;;N;;;;;
11BA7;MASARAM GONDI LETTER TTHA;Lo;0;L;;;;N;;;;;
```

```

11BA8;MASARAM GONDI LETTER DDA;Lo;0;L;;;;N;;;;;
11BA9;MASARAM GONDI LETTER DDHA;Lo;0;L;;;;N;;;;;
11BAA;MASARAM GONDI LETTER NNA;Lo;0;L;;;;N;;;;;
11BAB;MASARAM GONDI LETTER TA;Lo;0;L;;;;N;;;;;
11BAC;MASARAM GONDI LETTER THA;Lo;0;L;;;;N;;;;;
11BAD;MASARAM GONDI LETTER DA;Lo;0;L;;;;N;;;;;
11BAE;MASARAM GONDI LETTER DHA;Lo;0;L;;;;N;;;;;
11BAF;MASARAM GONDI LETTER NA;Lo;0;L;;;;N;;;;;
11BB0;MASARAM GONDI LETTER PA;Lo;0;L;;;;N;;;;;
11BB1;MASARAM GONDI LETTER PHA;Lo;0;L;;;;N;;;;;
11BB2;MASARAM GONDI LETTER BA;Lo;0;L;;;;N;;;;;
11BB3;MASARAM GONDI LETTER BHA;Lo;0;L;;;;N;;;;;
11BB4;MASARAM GONDI LETTER MA;Lo;0;L;;;;N;;;;;
11BB5;MASARAM GONDI LETTER YA;Lo;0;L;;;;N;;;;;
11BB6;MASARAM GONDI LETTER RA;Lo;0;L;;;;N;;;;;
11BB7;MASARAM GONDI LETTER LA;Lo;0;L;;;;N;;;;;
11BB8;MASARAM GONDI LETTER VA;Lo;0;L;;;;N;;;;;
11BB9;MASARAM GONDI LETTER SHA;Lo;0;L;;;;N;;;;;
11BBA;MASARAM GONDI LETTER SSA;Lo;0;L;;;;N;;;;;
11BBB;MASARAM GONDI LETTER SA;Lo;0;L;;;;N;;;;;
11BBC;MASARAM GONDI LETTER HA;Lo;0;L;;;;N;;;;;
11BBD;MASARAM GONDI LETTER LLA;Lo;0;L;;;;N;;;;;
11BBE;MASARAM GONDI LETTER KSSA;Lo;0;L;;;;N;;;;;
11BBF;MASARAM GONDI LETTER JNYA;Lo;0;L;;;;N;;;;;
11BC0;MASARAM GONDI LETTER TRA;Lo;0;L;;;;N;;;;;
11BC1;MASARAM GONDI VOWEL SIGN AA;Mn;0;NSM;;;;N;;;;;
11BC2;MASARAM GONDI VOWEL SIGN I;Mn;0;NSM;;;;N;;;;;
11BC3;MASARAM GONDI VOWEL SIGN II;Mn;0;NSM;;;;N;;;;;
11BC4;MASARAM GONDI VOWEL SIGN U;Mn;0;NSM;;;;N;;;;;
11BC5;MASARAM GONDI VOWEL SIGN UU;Mn;0;NSM;;;;N;;;;;
11BC6;MASARAM GONDI VOWEL SIGN VOCALIC R;Mn;0;NSM;;;;N;;;;;
11BCA;MASARAM GONDI VOWEL SIGN E;Mn;0;NSM;;;;N;;;;;
11BCC;MASARAM GONDI VOWEL SIGN AI;Mn;0;NSM;;;;N;;;;;
11BCD;MASARAM GONDI VOWEL SIGN O;Mn;0;NSM;;;;N;;;;;
11BCF;MASARAM GONDI VOWEL SIGN AU;Mn;0;NSM;;;;N;;;;;
11BD0;MASARAM GONDI SIGN ANUSVARA;Mn;0;NSM;;;;N;;;;;
11BD1;MASARAM GONDI SIGN VISARGA;Mn;0;NSM;;;;N;;;;;
11BD2;MASARAM GONDI SIGN NUKTA;Mn;7;NSM;;;;N;;;;;
11BD3;MASARAM GONDI SIGN CANDRA;Mn;0;NSM;;;;N;;;;;
11BD4;MASARAM GONDI SIGN HALANTA;Mn;0;NSM;;;;N;;;;;
11BD5;MASARAM GONDI VIRAMA;Mn;9;NSM;;;;N;;;;;
11BD6;MASARAM GONDI REPHA;Lo;0;L;;;;N;;;;;
11BD7;MASARAM GONDI RA-KARA;Mn;0;NSM;;;;N;;;;;
11BE0;MASARAM GONDI DIGIT ZERO;Nd;0;L;0;0;0;N;;;;;
11BE1;MASARAM GONDI DIGIT ONE;Nd;0;L;1;1;1;N;;;;;
11BE2;MASARAM GONDI DIGIT TWO;Nd;0;L;2;2;2;N;;;;;
11BE3;MASARAM GONDI DIGIT THREE;Nd;0;L;3;3;3;N;;;;;
11BE4;MASARAM GONDI DIGIT FOUR;Nd;0;L;4;4;4;N;;;;;
11BE5;MASARAM GONDI DIGIT FIVE;Nd;0;L;5;5;5;N;;;;;
11BE6;MASARAM GONDI DIGIT SIX;Nd;0;L;6;6;6;N;;;;;
11BE7;MASARAM GONDI DIGIT SEVEN;Nd;0;L;7;7;7;N;;;;;
11BE8;MASARAM GONDI DIGIT EIGHT;Nd;0;L;8;8;8;N;;;;;
11BE9;MASARAM GONDI DIGIT NINE;Nd;0;L;9;9;9;N;;;;;

```

5.2 Linebreaking

Linebreaking properties given in the data format of LineBreak.txt:

```

11B90..11BC0; AL # MASARAM GONDI LETTER A .. MASARAM GONDI LETTER TRA
11BC1..11BD4; CM # MASARAM GONDI SIGN AA .. MASARAM GONDI SIGN HALANTA

```

```

11BD5; CM          # MASARAM GONDI VIRAMA
11BC6; AL          # MASARAM GONDI REPHA
11BD7; CM          # MASARAM GONDI RA-KARA
11BE0..11BE9; NU   # MASARAM GONDI DIGIT ZERO .. MASARAM GONDI DIGIT NINE

```

5.3 Syllabic Categories

Syllabic categories given in the format of IndicSyllabicCategory.txt:

```

# Indic_Syllabic_Category=Bindu
11BD0          ; Bindu          # Mn          MASARAM GONDI SIGN ANUSVARA

# Indic_Syllabic_Category=Visarga
11BD1          ; Visarga        # Mc          MASARAM GONDI SIGN VISARGA

# Indic_Syllabic_Category=Nukta
11BD2          ; Nukta          # Mn          MASARAM GONDI SIGN NUKTA

# Indic_Syllabic_Category=Virama
11BD5          ; Virama         # Mn          MASARAM GONDI VIRAMA

# Indic_Syllabic_Category=Pure_Killer
11BD4          ; Pure_Killer    # Mn          MASARAM GONDI SIGN HALANTA

# Indic_Syllabic_Category=Vowel_Independent
11B90..11B9B   ; Vowel_Independent # Lo   [10] MASARAM GONDI LETTER A .. AU

# Indic_Syllabic_Category=Vowel_Dependent
11BC1..11BC6   ; Vowel_Dependent  # Mn   [6] MASARAM GONDI VOWEL SIGN AA .. VOCALIC R
11BCA          ; Vowel_Dependent  # Mn          MASARAM GONDI VOWEL SIGN E
11BCC..11BCD   ; Vowel_Dependent  # Mn   [2] MASARAM GONDI VOWEL SIGN AI .. O
11BCF          ; Vowel_Dependent  # Mn          MASARAM GONDI VOWEL SIGN AU
11BD3          ; Vowel_Dependent  # Mn          MASARAM GONDI SIGN CANDRA

# Indic_Syllabic_Category=Consonant
11B9C..11BC0   ; Consonant        # Lo   [40] MASARAM GONDI LETTER KA .. TRA

# Indic_Syllabic_Category=Consonant_Succeeding_Repha
11BD6          ; Consonant_Succeeding_Repha # Lo   MASARAM GONDI REPHA

# Indic_Syllabic_Category=Consonant_Subjoined
11BD7          ; Consonant_Subjoined # Mn          MASARAM GONDI RA-KARA

```

5.4 Positional Categories

Positional data for Gondi combining signs in the format of IndicPositionalCategory.txt:

```

# Indic_Positional_Category=Top
11BC1..11BC5   ; Top          # Mn   [5] MASARAM GONDI VOWEL SIGN AA .. UU
11BCA          ; Top          # Mn          MASARAM GONDI VOWEL SIGN E
11BCC..11BCD   ; Top          # Mn   [2] MASARAM GONDI VOWEL SIGN AI .. O
11BCF          ; Top          # Mn          MASARAM GONDI VOWEL SIGN AU
11BD0          ; Top          # Mn          MASARAM GONDI SIGN ANUSVARA
11BD1          ; Top          # Mn          MASARAM GONDI SIGN VISARGA
11BD3          ; Top          # Mn          MASARAM GONDI SIGN CANDRA

# Indic_Positional_Category=Bottom
11BC6          ; Bottom       # Mn          MASARAM GONDI VOWEL SIGN VOCALIC R

```

11BD2	; Bottom # Mn	MASARAM GONDI SIGN NUKTA
11BD4	; Bottom # Mn	MASARAM GONDI SIGN HALANTA
11BD7	; Bottom # Mn	MASARAM GONDI RA-KARA

5.5 ‘Confusable’ Characters

Gondi characters that bear resemblances to those of other scripts are listed below:

11BC1 MASARAM GONDI VOWEL SIGN AA	; 0304 COMBINING MACRON
11BB1 MASARAM GONDI LETTER PHA	; 1109D KAITHI LETTER NNA
11BBA MASARAM GONDI LETTER SSA	; 0398 GREEK CAPITAL LETTER THETA
11BE2 MASARAM GONDI DIGIT TWO	; 0055 LATIN CAPITAL LETTER U

6 References

- Anderson, Deborah; et al. 2015. “Recommendations to UTC #142 February 2015 on Script Proposals” (L2/15-045). <http://www.unicode.org/L2/L2015/15045-script-rec.pdf>
- गुरुजी, मनीराम दुर्गा [Gurūjī, Manīrāma Durgā]. गोंदी लम्क पुंदान [Gomḍī lamka puṁdāna].
- मण्डाले, सीताराम [Maṇḍāle, Sītārām]. कोयाबोली [Koyābolī]. गोंदी शब्द संग्रह - गोंदी, मराठी, हिन्दी [Gomḍī Śabda Saṁgraha - Gomḍī, Marāṭhī, Hindī].
- Masaram, Bhava Singh. 1951. “गोंदी लिपि” [Gomḍī lipi]. Central Institute of Indian Languages, Multimedia library, photograph no. 64.
- Pandey, Anshuman. 2010. “Preliminary Proposal to Encode the Gondi Script in the UCS” (L2/10-207). <http://www.unicode.org/L2/L2010/10207-gondi.pdf>
- . 2012. “Revised Preliminary Proposal to Encode the Gondi Script” (L2/12-235). <http://www.unicode.org/L2/L2012/12235-n4291-gondi.pdf>
- . 2015a. “Proposal to Encode the Gondi Script” (L2/15-005). <http://www.unicode.org/L2/L2015/15005-gondi.pdf>
- . 2015b. “Preliminary Proposal to Encode the Gunjala Gondi Script in Unicode” (L2/15-086). <http://www.unicode.org/L2/L2015/15086-gunjala-gondi.pdf>
- . 2015c. “Proposal to Encode the ‘Parsapen’ Symbol in Unicode” (L2/15-111) <http://www.unicode.org/L2/L2015/15111-parsapen-symbol.pdf>
- Ramakrishna, G., N. Gayathri, Debiprasad Chattopadhyaya. 1983. *An Encyclopaedia of South Indian Culture*. Calcutta: K. P. Bagchi & Co.
- रामानन्द [Rāmānanda]. गोंदी अक्षर ज्ञान [Gomḍī Akṣara Jñāna].
- Vahia, M. N.; Ganesh Halkare. 2013. “Aspects of Gondi Astronomy”. *Journal of Astronomical History and Heritage*, vol. 16, no. 1, pp. 29–44. <http://www.tifr.res.in/~archaeo/VahiaGondTrip/JAHHGondsfinal.pdf>

7 Acknowledgments

I am deeply indebted to Dr. Mukund Gokhale (Script Research Institute, Pune) for providing me with numerous materials in Masaram's Gondi script and for patiently answering my numerous questions about the script over the past few years. Dr. Gokhale's efforts in developing typefaces for this Gondi script have helped to address issues regarding the interaction of multiple glyphs with a base letter. I also appreciate him for contacting Dr. Motiravan Kangle (Akhil-Gondwana Gondi Sahitya Parishad, Nagpur) with my questions. I thank Dr. Kangle for indulging my inquiries and for providing examples of current orthography of *ra* in conjuncts, and the usage of *halanta* and *candra*. The Gondi chart shown in figure ?? was provided by B. A. Sharada and Suman Kumari of the Central Institute of Indian Languages (Mysore). Mark Penny provided the chart shown in figure 6. I am also thankful to Roozbeh Pournader (Google) and Andrew Glass (Microsoft) for sharing their insights regarding effective models for representing the Gondi RA in conjuncts.

This project was made possible in part through a Google Research Award, granted to Deborah Anderson for the Script Encoding Initiative, and a grant from the United States National Endowment for the Humanities (PR-50205-15), which funds the Universal Scripts Project (part of the Script Encoding Initiative at the University of California, Berkeley). Any views, findings, conclusions or recommendations expressed in this publication do not necessarily reflect those of Google or the National Endowment for the Humanities.

Copyright © 2015 Anshuman Pandey. All rights reserved.

	11B9	11BA	11BB	11BC	11BD	11BE
0	 11B90	 11BA0	 11BB0	 11BC0	 11BD0	 11BE0
1	 11B91	 11BA1	 11BB1	 11BC1	 11BD1	 11BE1
2	 11B92	 11BA2	 11BB2	 11BC2	 11BD2	 11BE2
3	 11B93	 11BA3	 11BB3	 11BC3	 11BD3	 11BE3
4	 11B94	 11BA4	 11BB4	 11BC4	 11BD4	 11BE4
5	 11B95	 11BA5	 11BB5	 11BC5	 11BD5	 11BE5
6	 11B96	 11BA6	 11BB6	 11BC6	 11BD6	 11BE6
7	 11B97	 11BA7	 11BB7	 11BC7	 11BD7	 11BE7
8	 11B98	 11BA8	 11BB8	 11BC8	 11BD8	 11BE8
9	 11B99	 11BA9	 11BB9	 11BC9	 11BD9	 11BE9
A	 11B9A	 11BAA	 11BBA	 11BCA	 11BDA	 11BEA
B	 11B9B	 11BAB	 11BBB	 11BCB	 11BDB	 11BEB
C	 11B9C	 11BAC	 11BBC	 11BCC	 11BCD	 11BED
D	 11B9D	 11BAD	 11BBD	 11BCD	 11BDE	 11BEE
E	 11B9E	 11BAE	 11BBE	 11BCE	 11BDE	 11BEE
F	 11B9F	 11BAF	 11BBF	 11BCF	 11BDF	 11BEF

Vowels

11B90	𑌒	MASARAM GONDI LETTER A
11B91	𑌓	MASARAM GONDI LETTER AA
11B92	𑌔	MASARAM GONDI LETTER I
11B93	𑌕	MASARAM GONDI LETTER II
11B94	𑌖	MASARAM GONDI LETTER U
11B95	𑌗	MASARAM GONDI LETTER UU
11B96	𑌘	MASARAM GONDI LETTER E
11B97	𑌙	<reserved>
11B98	𑌚	MASARAM GONDI LETTER AI
11B99	𑌛	MASARAM GONDI LETTER O
11B9A	𑌜	<reserved>
11B9B	𑌝	MASARAM GONDI LETTER AU

Consonants

11B9C	𑌞	MASARAM GONDI LETTER KA
11B9D	𑌟	MASARAM GONDI LETTER KHA
11B9E	𑌠	MASARAM GONDI LETTER GA
11B9F	𑌡	MASARAM GONDI LETTER GHA
11BA0	𑌢	MASARAM GONDI LETTER NGA
11BA1	𑌣	MASARAM GONDI LETTER CA
11BA2	𑌤	MASARAM GONDI LETTER CHA
11BA3	𑌥	MASARAM GONDI LETTER JA
11BA4	𑌦	MASARAM GONDI LETTER JHA
11BA5	𑌧	MASARAM GONDI LETTER NYA
11BA6	𑌨	MASARAM GONDI LETTER TTA
11BA7	𑌩	MASARAM GONDI LETTER TTHA
11BA8	𑌪	MASARAM GONDI LETTER DDA
11BA9	𑌫	MASARAM GONDI LETTER DDHA
11BAA	𑌬	MASARAM GONDI LETTER NNA
11BAB	𑌭	MASARAM GONDI LETTER TA
11BAC	𑌮	MASARAM GONDI LETTER THA
11BAD	𑌯	MASARAM GONDI LETTER DA
11BAE	𑌰	MASARAM GONDI LETTER DHA
11BAF	𑌱	MASARAM GONDI LETTER NA
11BB0	𑌲	MASARAM GONDI LETTER PA
11BB1	𑌳	MASARAM GONDI LETTER PHA
11BB2	𑌴	MASARAM GONDI LETTER BA
11BB3	𑌵	MASARAM GONDI LETTER BHA
11BB4	𑌶	MASARAM GONDI LETTER MA
11BB5	𑌷	MASARAM GONDI LETTER YA
11BB6	𑌸	MASARAM GONDI LETTER RA
11BB7	𑌹	MASARAM GONDI LETTER LA
11BB8	𑌺	MASARAM GONDI LETTER VA
11BB9	𑌻	MASARAM GONDI LETTER SHA
11BBA	𑌼	MASARAM GONDI LETTER SSA
11BBB	𑌽	MASARAM GONDI LETTER SA
11BBC	𑌾	MASARAM GONDI LETTER HA
11BBD	𑌿	MASARAM GONDI LETTER LLA

Conjunct letters

11BBE	𑌿𑌞	MASARAM GONDI LETTER KSSA
11BBF	𑌿𑌟	MASARAM GONDI LETTER JNYA
11BC0	𑌿𑌠	MASARAM GONDI LETTER TRA

Dependent vowel signs

11BC1	𑌿𑌒	MASARAM GONDI VOWEL SIGN AA
11BC2	𑌿𑌓	MASARAM GONDI VOWEL SIGN I
11BC3	𑌿𑌔	MASARAM GONDI VOWEL SIGN II
11BC4	𑌿𑌕	MASARAM GONDI VOWEL SIGN U
11BC5	𑌿𑌖	MASARAM GONDI VOWEL SIGN UU
11BC6	𑌿𑌗	MASARAM GONDI VOWEL SIGN VOCALIC R
11BC7	𑌿𑌘	<reserved>
11BC8	𑌿𑌙	<reserved>
11BC9	𑌿𑌚	<reserved>
11BCA	𑌿𑌛	MASARAM GONDI VOWEL SIGN E
11BCB	𑌿𑌜	<reserved>

11BCC	𑌿𑌝	MASARAM GONDI VOWEL SIGN AI
11BCD	𑌿𑌞	MASARAM GONDI VOWEL SIGN O
11BCE	𑌿𑌟	<reserved>
11BCF	𑌿𑌠	MASARAM GONDI VOWEL SIGN AU

Various signs

11BD0	𑌿𑌒	MASARAM GONDI SIGN ANUSVARA
11BD1	𑌿𑌓	MASARAM GONDI SIGN VISARGA
11BD2	𑌿𑌔	MASARAM GONDI SIGN NUKTA
11BD3	𑌿𑌕	MASARAM GONDI SIGN CANDRA
		• used for transcribing foreign vowels
11BD4	𑌿𑌖	MASARAM GONDI SIGN HALANTA
		• used for silencing the inherent vowel

Virama

11BD5	𑌿𑌗	MASARAM GONDI VIRAMA
		• used for producing conjuncts

Cluster-specific consonant forms

11BD6	𑌿𑌟	MASARAM GONDI REPHA
		• cluster-initial form of RA
11BD7	𑌿𑌠	MASARAM GONDI RA-KARA
		• cluster-final form of RA

Digits

11BE0	0	MASARAM GONDI DIGIT ZERO
11BE1	𑌒	MASARAM GONDI DIGIT ONE
11BE2	𑌓	MASARAM GONDI DIGIT TWO
11BE3	𑌔	MASARAM GONDI DIGIT THREE
11BE4	𑌕	MASARAM GONDI DIGIT FOUR
11BE5	𑌖	MASARAM GONDI DIGIT FIVE
11BE6	𑌗	MASARAM GONDI DIGIT SIX
11BE7	𑌘	MASARAM GONDI DIGIT SEVEN
11BE8	𑌙	MASARAM GONDI DIGIT EIGHT
11BE9	𑌚	MASARAM GONDI DIGIT NINE



Figure 1: A document illustrating the characters and basic principles of Mangal Singh Masaram’s Gondi script. It was created by Mangal Singh’s son Bhava Singh in 1951. Image courtesy of the Central Institute of Indian Languages, Mysore.

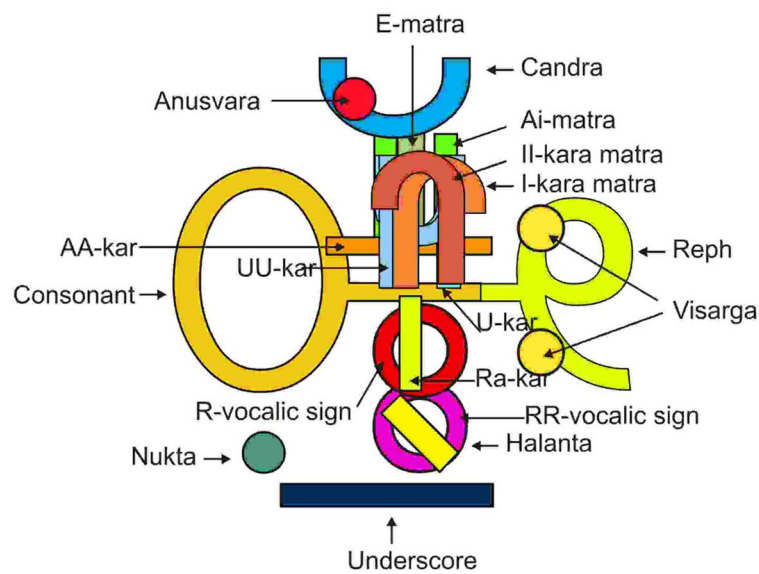
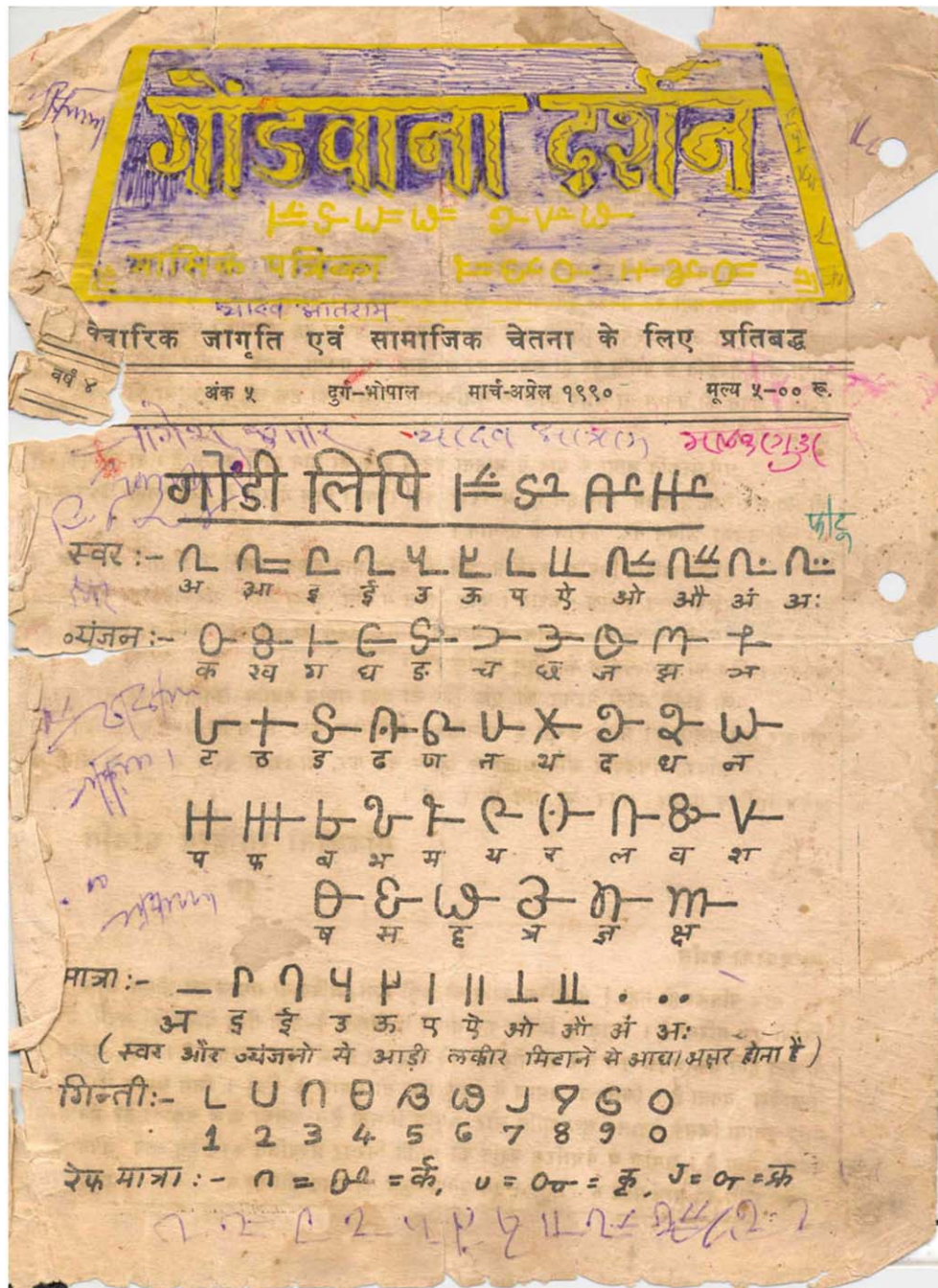


Figure 2: A schematic designed by Mukund Gokhale showing the position of all combining signs on a base letter. The diagram shows a sign labelled “RR-vocalic sign”, which is a length variant of  VOWEL SIGN VOCALIC R introduced by Gokhale (see section 4.11).

Figure 3: Cover of *Gondwana Darshan* (March-April 1990, vol. 5).

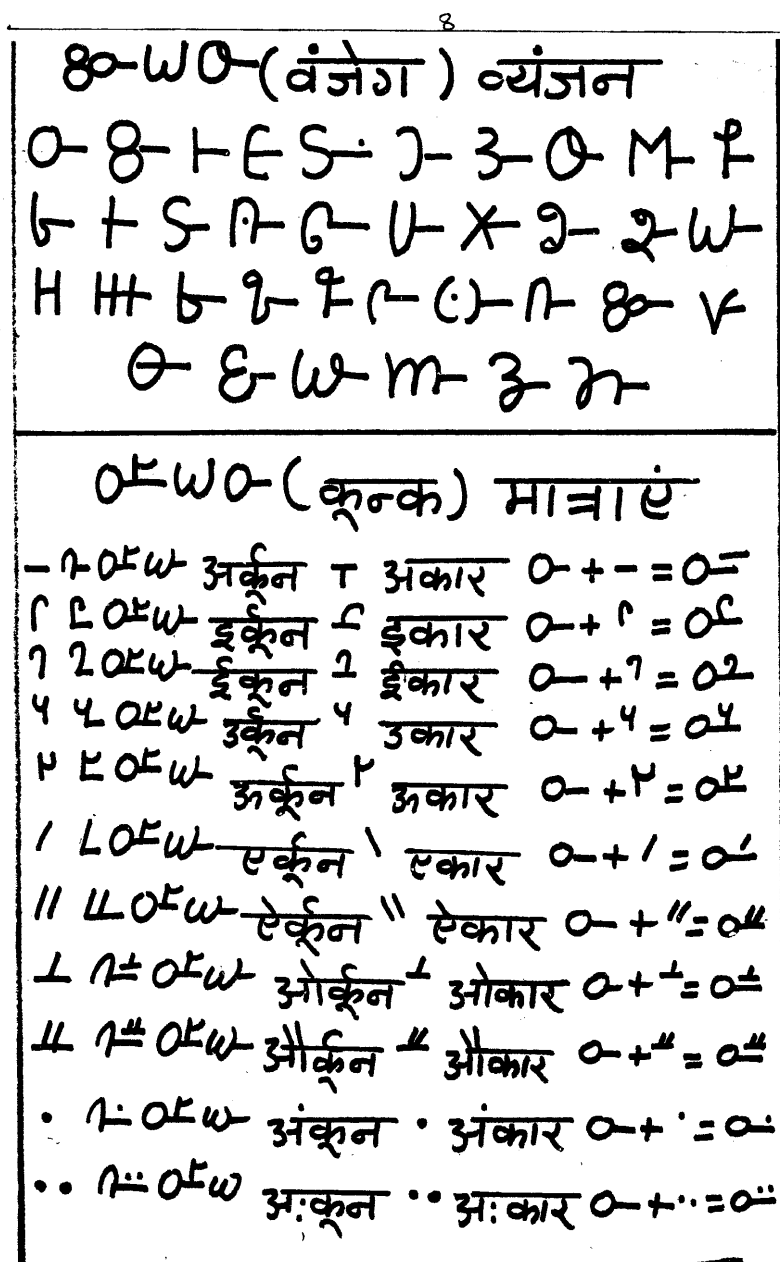


Figure 4: Page from *Gomḍī Akṣara Jñāna* showing consonants and vowel signs, and consonant-vowel combinations (from Rāmānanda: 8).

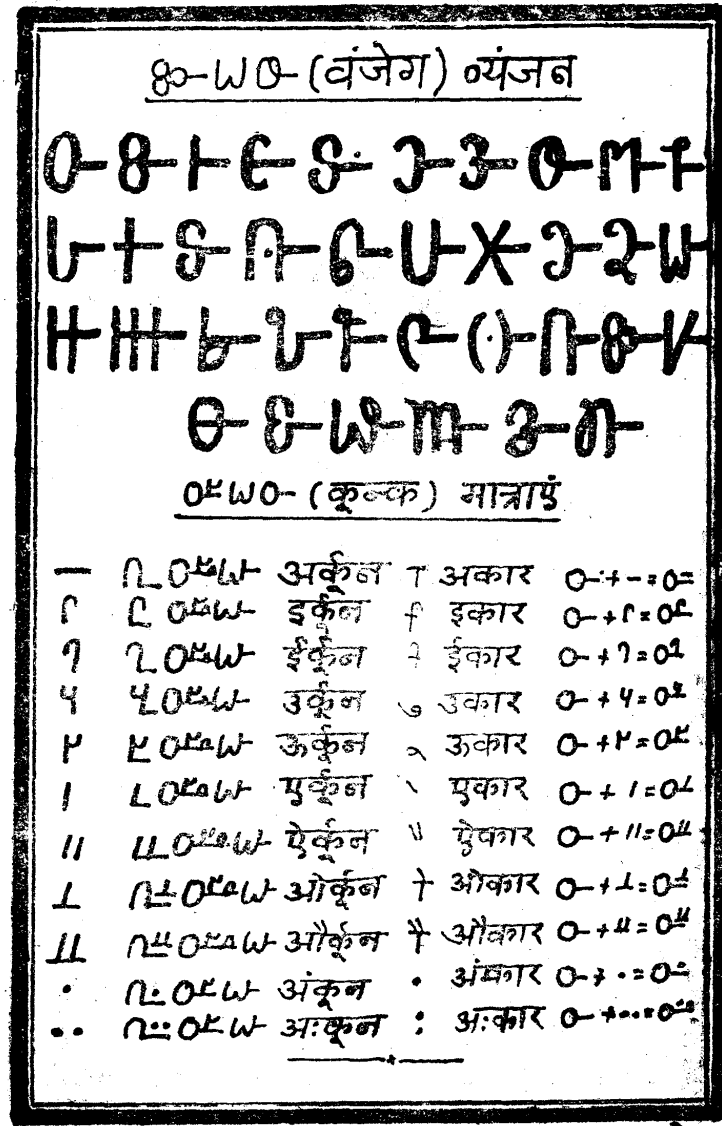


Figure 5: Page from *Gomḍī Lamk Pundan* showing consonants and vowel signs, and consonant-vowel combinations (from Gurujī: 11).

Consonants व्यञ्जन

	Cons. only	– [voice]		+ [voice]		Nasals	Cons. with word
<i>Aspiration</i>		– [asp]	+ [asp]	– [asp]	+ [asp]		
Velar		ᳵ	ᳶ	᳷	᳸	᳹	
Palatal		ᳺ	᳻	᳼	᳽	᳾	
Retroflex		᳽	᳾	᳿	ᳺ	᳻	
Dental		᳼	᳽	᳾	᳿	ᳺ	
Bilabial		ᳺ	᳻	᳼	᳽	᳾	
Liquids and Semi-Vowels		᳽	᳾	᳿	ᳺ	᳻	
Fricatives		᳽	᳾	᳿	ᳺ	᳻	
Affricates			᳽	᳾	᳿		

Vowels स्वर

	ᳵ	ᳶ	᳷	᳸	᳹		
	ᳺ	᳻	᳼	᳽	᳾	᳿	

Figure 6: A handwritten chart of the Gondi script. Source: Ramesh Gedam and Mark Penny (2001).

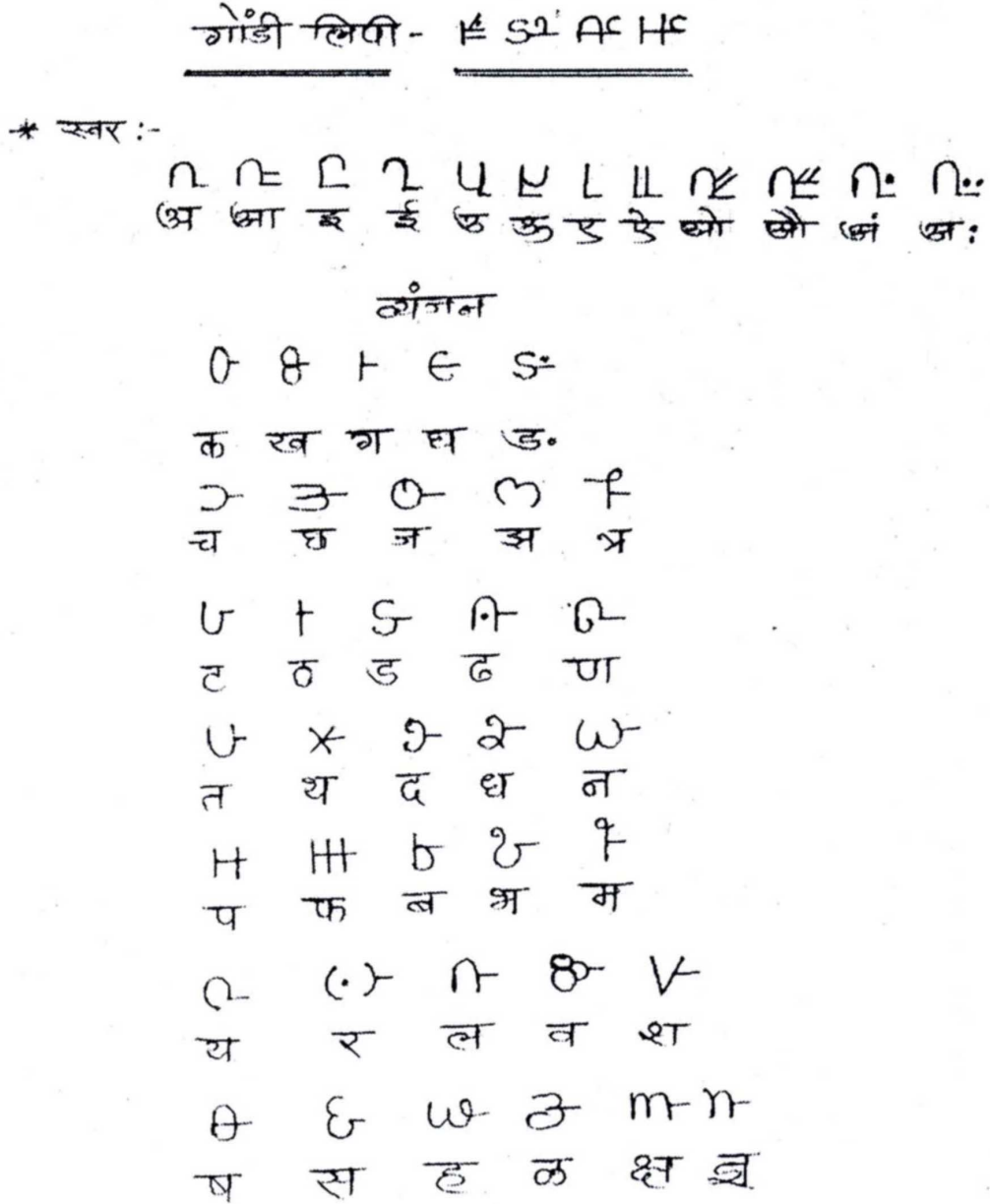


Figure 7: A handwritten chart of the Gondi script (Maṇḍāle 2008: 8).

अ		२५-०= अटका २५= अना मदका
आ		२५-०= आकी २५-५= आमाट पत्ती
इ		२५-०= इमला २५= इमा मकान
ई		२५-०= ईली २५-५= इमाट इली
उ		५६-०= उसकाळ ५६= उद मूसळ
ऊ		५६-०= ऊल्लू ५६-५= ऊबक च्याज
ऋ		५६-०= ऋली ५६-५= ऋर बकरी

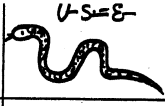
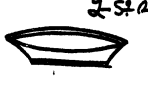
ॐ		५६-०= ॐनक ५६-५= ॐक चडमा
अं		५६-०= अंटा ५६-५= अंर वहलान
अः		५६-०= अंजार ५६-५= अंकर हथियार
अं		५६-०= अंबाडी ५६-५= अंकर सबजी-भाजी
अः		५६-०= अः ५६-५= अः अः

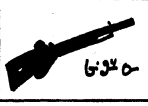
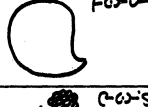
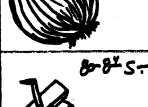
ॐ ॐ सर्क (स्वर)
५६-०= ॐ ॐ ॐ ॐ ॐ ॐ ॐ ॐ ॐ ॐ
अं आ इ ई उ ऊ ऋ ॐ ॐ
अं अः
अं अः

क		०-०-०= कळी ०-०-५= कचरा कान
ख		०-०-५= खड्ड ०-०-५= खटारा खाट
ग		०-०-५= गडुचा ०-०-५= गवरा आसन
घ		०-०-५= घट्टा ०-०-५= घन घन्टी
ङ		०-०-५= अडा ०-०-५= नडा ङ
च		०-०-५= चकरी ०-०-५= चराट चक्र
छ		०-०-५= छली डी ०-०-५= छपरी छाता

ज		०-०-५= जल ०-०-५= जलान
झ		०-०-५= झंडा ०-०-५= झपाटा
ञ		०-०-५= अमरा ०-०-५= अमोता
ट		०-०-५= टबडी ०-०-५= टरबुज
ठ		०-०-५= ठप्पा ०-०-५= ठाना
ड		०-०-५= डबरु ०-०-५= डब्बा
ढ		०-०-५= ढाठा ०-०-५= ढमढम

Figure 8: Page from *Gomḍī Akṣara Jñāna* showing vowel letters (from Rāmānanda: 1–4).

त		५-५=६- तडास ५-१०=०- नाज तडाबार
थ		५-१२ थानी ५-५१ थड़ी छोटा
द		३-०-५१ दवड़ी ३-०=१ दबाई डलियाँ
ध		२-५१-१२ धडीया २-००= धक्का कोपर
न		५५२- नय ५५५५५५५५ कुना नक्कर
प		५५५५५५५५ पन्ने ५५५५५५५५ मेदक
फ		५५५५५५५५ फन्नी ५५५५५५५५ फनकाड़ फन

ब		६-३५० बन्दूक ६-०५५ बरमो बन्दूक
भ		१-१५ भाला १-०५ भर्त बरही
म		१-०५०= मरका १-५= मड़ा आम
य		१-०५५५५५५५ येरन्डी १-०५५५५५५५ आरणी
र		०५५५५५५५५५ रन्दा ०५५५५५५५५५ रिन्दा
ल		१-०५५५५५५५५५ लसून १-०५५५५५५५५५ लहसून
व		०५५५५५५५५५५५ वरबुड़ ०५५५५५५५५५५५ बरबर

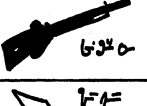
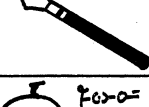
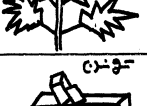
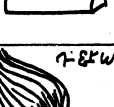
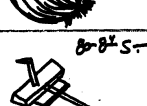
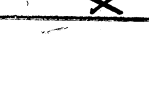
ब		६-३५० बन्दूक ६-०५५ बरमो बन्दूक
भ		१-१५ भाला १-०५ भर्त बरही
म		१-०५०= मरका १-५= मड़ा आम
य		१-०५५५५५५५५५ येरन्डी १-०५५५५५५५५५ आरणी
र		०५५५५५५५५५५५ रन्दा ०५५५५५५५५५५५ रिन्दा
ल		१-०५५५५५५५५५५५ लसून १-०५५५५५५५५५५५ लहसून
व		०५५५५५५५५५५५५५ वरबुड़ ०५५५५५५५५५५५५५ बरबर

Figure 9: Page from *Gomḍī Akṣara Jñāna* showing the letters TA .. VA (from Rāmānanda: 5–7).

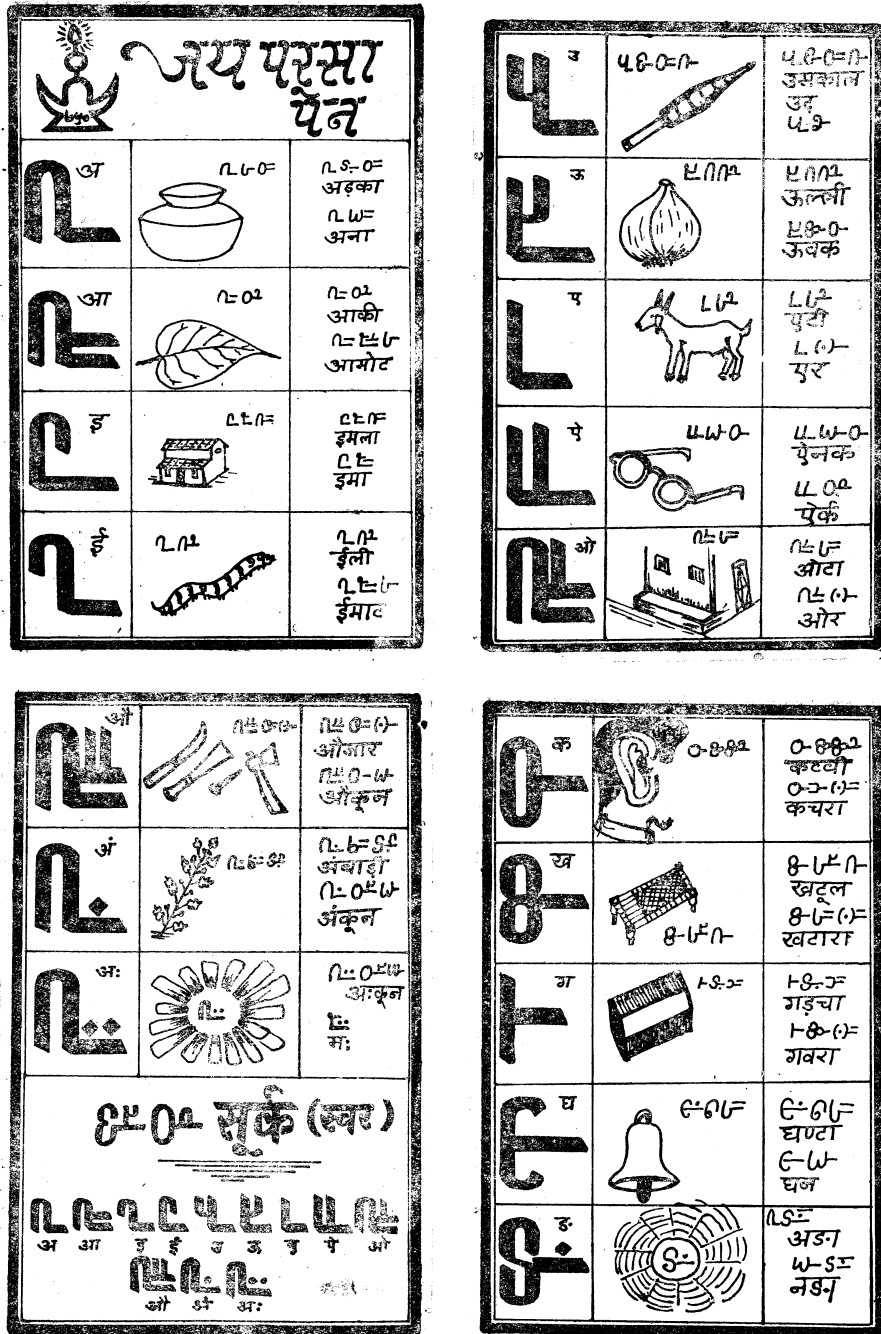
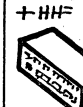
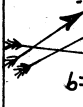


Figure 10: Page from *Gomḍī Lamk Pundan* showing vowel letters (from Guruji: 1–4). The first page (top right corner) shows the *persapen* symbol.

च		३०-१३ ३-१-१ यराट
छ		३-१५-३३ छलोडी ३-१५-३३ छपरी
ज		७-११-७ जला ७-११-७ जन्नाग
झ		१५-३ झडा १५-३ झपाटा
ञ		१-१-१ अमरा १-१-१ अमोता

ट		१-७-३३ टवडी १-७-३३ टरबुज
ठ		१-११-७ ठप्पा १-११-७ ठाना
ड		७-७-१५ डबर ७-७-१५ डबा
ढ		१-११-१३ ढपली १-११-१३ ढमढम
ण		७-७-७ बाण १-१-७ गण

त		१-७-७ तडास १-७-७ तलवार
थ		१-११-७ थाली १-११-७ थडी
द		७-७-३३ दवडी ७-७-३३ दवाई
ध		३-३३-७ धडीया ३-३३-७ धक्का
न		१-१२ नय १-१२ नत्तर

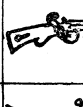
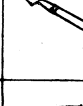
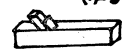
प		१-११-७ पन्ने १-११-७ पलंग
फ		११-११-७ फन्नी ११-११-७ फनकाड
ब		७-७-३३ बदुक ७-७-३३ बरमा
भ		१-११-७ भाला १-११-७ भरु
म		१-११-७ मरका १-११-७ मडा

Figure 11: Page from *Gomḍi Lamk Pundan* showing vowel letters (from Guruji: 5–8).

य		२-१-२ यरंडी २-२-१ आयाल
र		१-२-२ रंदा १-२-२ रच्छा
ल		१-२-२ लसून १-०-२ लंका
व		२-२-२ खरुड २-१-२ वरचे
श		१-२-० शरूक १-२-२ शम्मे

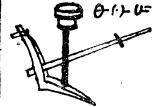
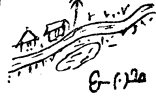
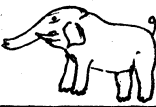
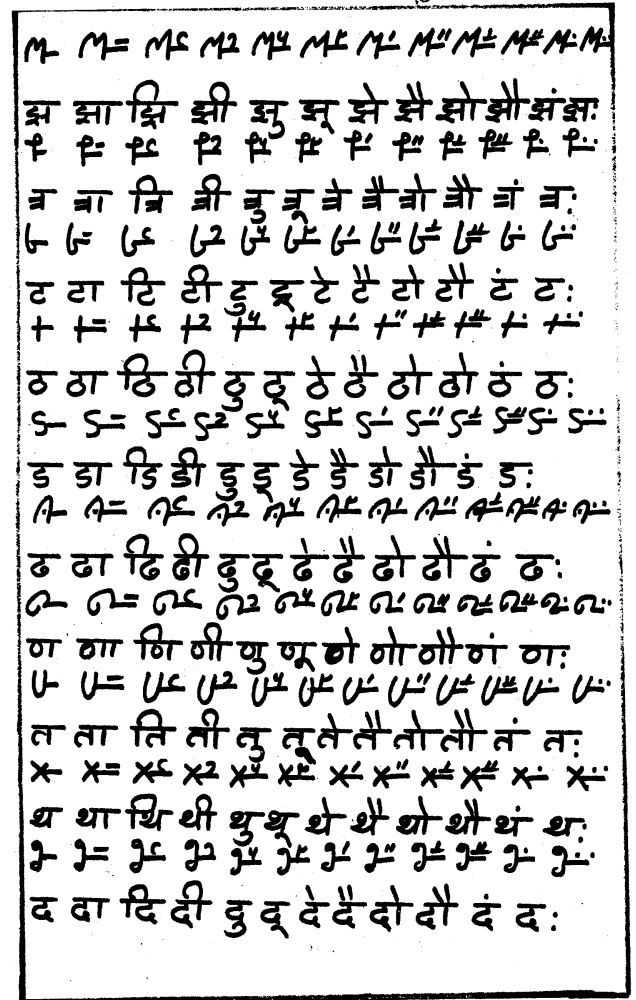
ष		२-१-२ बरता २-१-२ षटार
स		२-१-२ सररी २-२-२ सबोड
ह		२-२-२ हल्ली २-२-२ हर्नाल
क्ष		२-२-२ क्षणी २-२-२ क्षनी
त्र		२-२-२ त्रपाल
ज्ञ		२-२-२ ज्ञान

Figure 12: Page from *Gomḍī Lamk Pundan* showing vowel letters (from Guruji: 9–10).



33

[illegible]

34

Figure 15: Page from *Gomḍī Akṣara Jñāna* explaining conjunct formation (from Rāmānanda: 13). Bottom half describes the usage of Latin marks of punctuation.

H S=I (पाड़ांग) गिनती		
L उन्दी एक	W सांरंग छह	
U रंड दो	J येरंग सात	
n मूंद तीन	9 अरंग आठ	
θ नालुंग चार	6 नरंग नौ	
β सयुंग पांच	LO पद दस	
नुर बेरी पाड़ांग (सौ तक गिनती)		
L उन्दी	LL पददी	UL रण्डुन्दी
U रंड	LU पदंड	UU रण्डुन्ड
n मूंद	Ln पदुंद	Un रण्डुन्त
θ नालुंग	Lθ पनालू	Uθ रणनालू
β सयुंग	Lβ पसयुं	Uβ रन्सयुं
W सांरंग	LW पसारु	UW रन्सारु
J येरंग	LJ पदेरु	UJ रन्डेरु
9 अरंग	L9 पदारु	U9 रन्डारु
6 नरंग	L6 पनरु	U6 रनरु
LO पद	UO रण्डू	NO मण्डू

Figure 16: Page from *Gomḍī Akṣara Jñāna* showing Gondi digits (from Rāmānanda: 14).

H S=I (पाड़ांग) गिनती		
L उन्दी एक	W सांरंग छह	
U रंड दो	J येरंग सात	
n मूंद तीन	9 अरंग आठ	
θ नालुंग चार	6 नरंग नव	
β सयुंग पांच	LO पद दस	
नुर बेरी पाड़ांग (सौ तक गिनती)		
L उन्दी	LL पदुंदी	UL रण्डुन्दी
U रंड	LU पदंड	UU रण्डुन्ड
n मूंद	Ln पदुंद	Un रण्डुन्त
θ नालुंग	Lθ पनालू	Uθ रणनालू
β सयुंग	Lβ पसयुं	Uβ रन्सयुं
W सांरंग	LW पसारु	UW रन्सारु
J येरंग	LJ पदेरु	UJ रन्डेरु
9 अरंग	L9 पदारु	U9 रन्डारु
6 नरंग	L6 पनरु	U6 रनरु
LO पद	UO रंडू	NO मण्डू

Figure 17: Page from *Gomḍī Lamk Pundan* showing Gondi digits (from Guruji: 14).

Number	Name	Style in Maharashtra	Style in Madhya Pradesh
1	Undi	𑌀	𑌀
2	Rand	𑌁	𑌁
3	Munda	𑌂	𑌂
4	Nalung	𑌃	𑌃
5	Sayyung	𑌄	𑌄
6	Sarung	𑌅	𑌅
7	Yerung	𑌆	𑌆
8	Arung	𑌇	𑌇
9	Narung	𑌈	𑌈
10	Pad	𑌉	𑌉

Figure 18: Comparison of regional variants of Gondi digits (from Vahia and Halkare 2013: 33).

[illegible]

Figure 19: A document showing usage of `CO-LLA`.

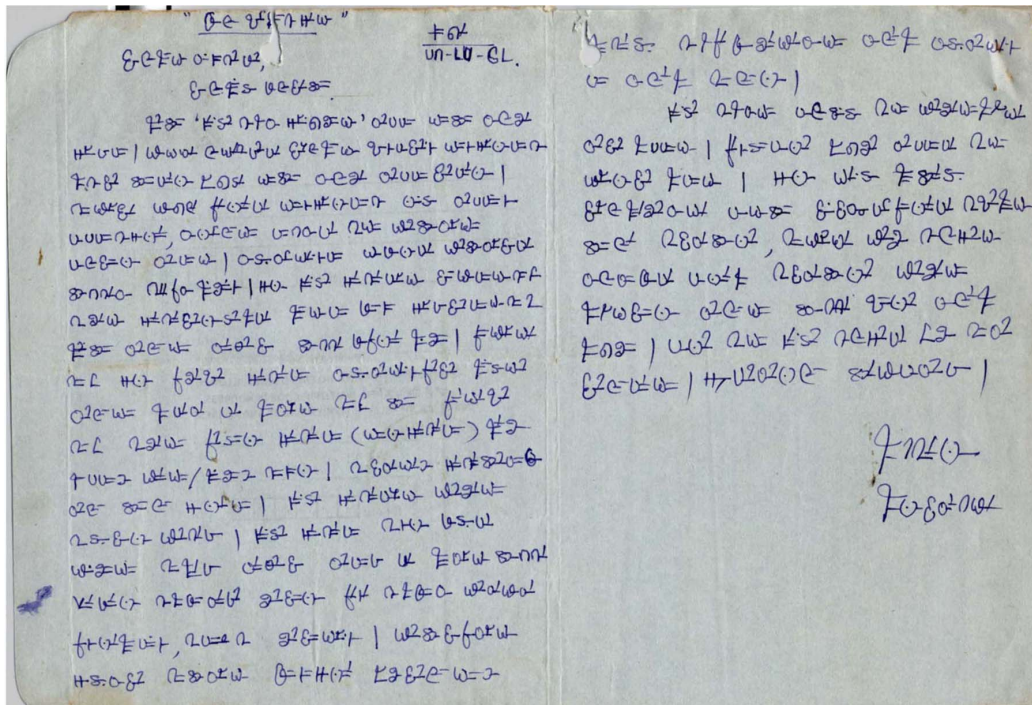


Figure 20: A letter written in Gondi.

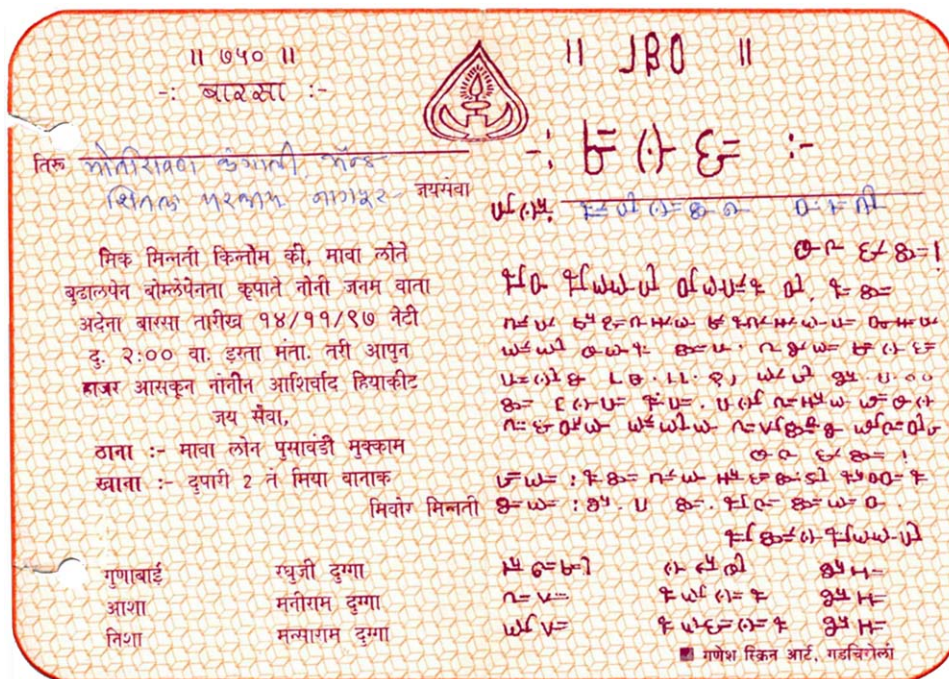


Figure 21: An invitation card written in the Devanagari and Gondi scripts. The persapen symbol appears in the top center of the card.

$$H^1\mathcal{O}(-E) = H^1\mathcal{O}(-H) = H^4\mathcal{O}(-2H) = H^6 = 0$$
[illegible]

ፍጋፍ፡፡ ፍጋፍ፡፡ (ፍጋፍ፡፡)

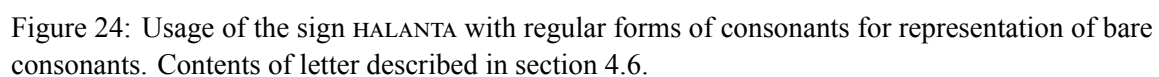
ጊዜያዊነ ሁሉም ጉዳዮች ስላሉ፣ ዘጠኙ.

40

Text is complete "Vande Mataram"
This 'Gondi' Script font is designed by : Prof Mukund Gokhale
Script Research Institute
mwgokhale_scriptresearch@yahoo.co.in
gokhale.mukund1@gmail.com

Dedicated to the inventor of the script 'Munshi Mangal Singh Masaram'

41



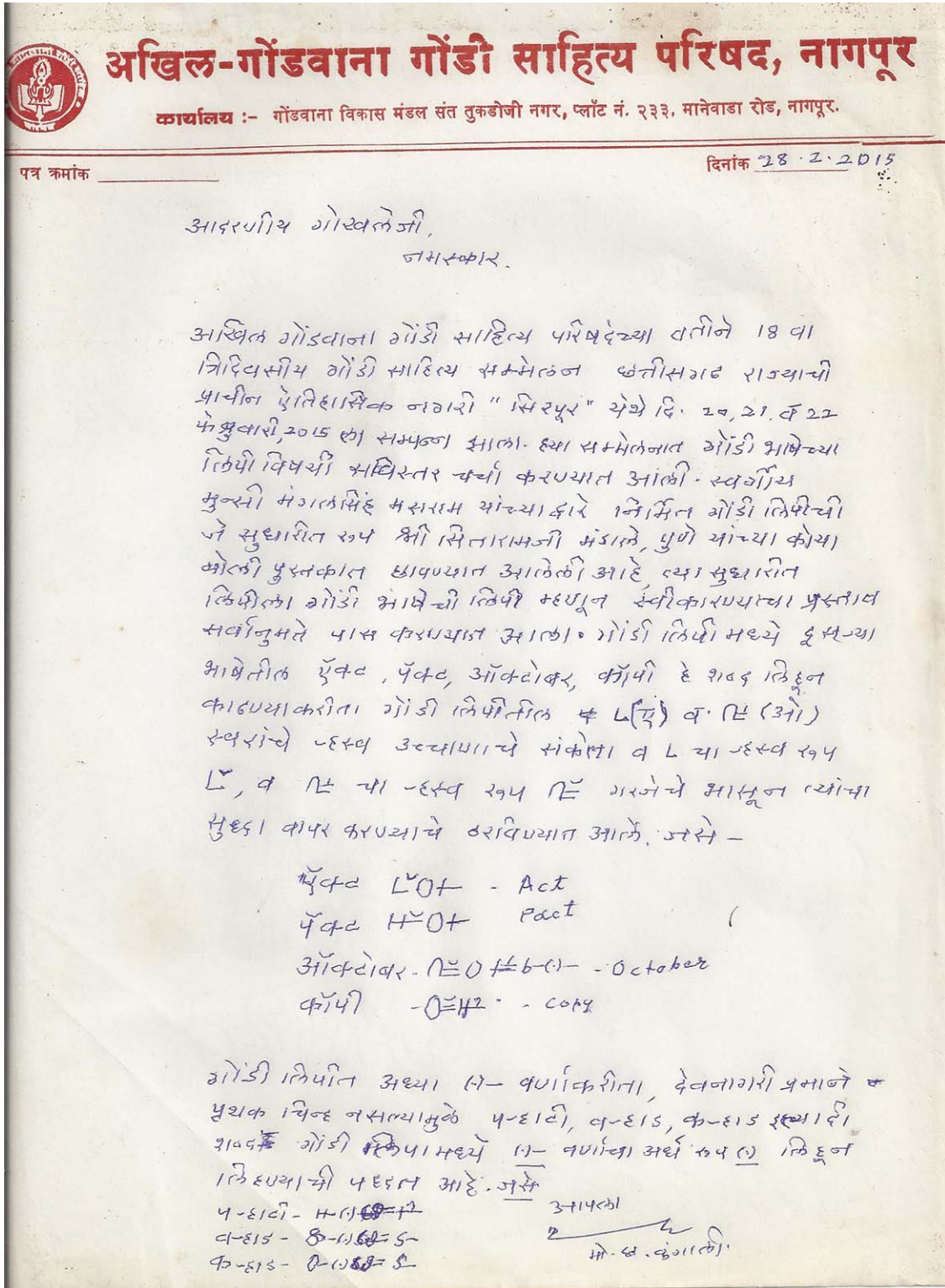


Figure 25: Usage of the sign CANDRA. Contents of letter described in section 4.4.



Figure 26: A calendar showing names for days of the week in Gondi and dates in Gondi digits.

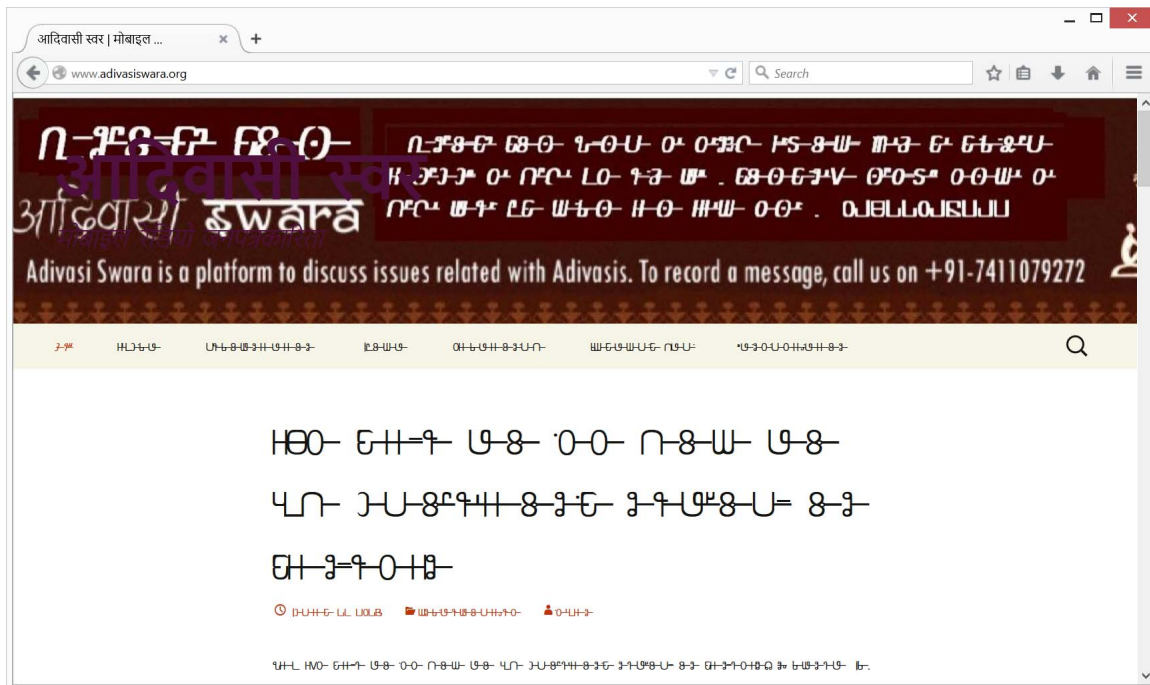


Figure 27: Screen-shot of adivasiswara.org showing content in Gondri script. The text in the banner is valid Gondri content. The text in the main frame of the site, however, is invalid: the script representing meaningless sequences of Gondri letters as the underlying text is Latin-script content in English. The Gondri text is represented using a server-side font based upon the Latin encoding that contains Gondri glyphs. The site appears to be a work-in-progress and is awaiting proper support for the Gondri script in Unicode in order to deliver proper content.

ISO/IEC JTC 1/SC 2/WG 2
PROPOSAL SUMMARY FORM TO ACCOMPANY SUBMISSIONS
FOR ADDITIONS TO THE REPERTOIRE OF ISO/IEC 10646¹

Please fill all the sections A, B and C below.

Please read Principles and Procedures Document (P & P) from <http://www.dkuug.dk/JTC1/SC2/WG2/docs/principles.html> for guidelines and details before filling this form.

Please ensure you are using the latest Form from <http://www.dkuug.dk/JTC1/SC2/WG2/docs/summaryform.html>.

See also <http://www.dkuug.dk/JTC1/SC2/WG2/docs/roadmaps.html> for latest Roadmaps.

A. Administrative

1. Title:	Proposal to Encode the Masaram Gondi Script in Unicode		
2. Requester's name:	<i>Script Encoding Initiative (SEI) / Anshuman Pandey (anshuman.pandey@berkeley.edu)</i>		
3. Requester type (Member body/Liaison/Individual contribution):	<i>Liaison contribution</i>		
4. Submission date:	<i>2015-04-15</i>		
5. Requester's reference (if applicable):			
6. Choose one of the following:			
This is a complete proposal:			<i>Yes</i>
(or) More information will be provided later:			

B. Technical – General

1. Choose one of the following:			
a. This proposal is for a new script (set of characters):			<i>Yes</i>
Proposed name of script:	<i>Masaram Gondi</i>		
b. The proposal is for addition of character(s) to an existing block:			
Name of the existing block:			
2. Number of characters in proposal:			<i>75</i>
3. Proposed category (select one from below - see section 2.2 of P&P document):			
A-Contemporary ☒	B.1-Specialized (small collection)	B.2-Specialized (large collection)	
C-Major extinct ☐	D-Attested extinct	E-Minor extinct	
F-Archaic Hieroglyphic or Ideographic ☐	G-Obscure or questionable usage symbols		
4. Is a repertoire including character names provided?			<i>Yes</i>
a. If YES, are the names in accordance with the "character naming guidelines" in Annex L of P&P document?			<i>Yes</i>
b. Are the character shapes attached in a legible form suitable for review?			<i>Yes</i>
5. Fonts related:			
a. Who will provide the appropriate computerized font to the Project Editor of 10646 for publishing the standard?	<i>Anshuman Pandey</i>		
b. Identify the party granting a license for use of the font by the editors (include address, e-mail, ftp-site, etc.):	<i>Anshuman Pandey (anshuman.pandey@berkeley.edu)</i>		
6. References:			
a. Are references (to other character sets, dictionaries, descriptive texts etc.) provided?			<i>Yes</i>
b. Are published examples of use (such as samples from newspapers, magazines, or other sources) of proposed characters attached?			<i>Yes</i>
7. Special encoding issues:			
Does the proposal address other aspects of character data processing (if applicable) such as input, presentation, sorting, searching, indexing, transliteration etc. (if yes please enclose information)?			<i>Yes</i>

8. Additional Information:

Submitters are invited to provide any additional information about Properties of the proposed Character(s) or Script that will assist in correct understanding of and correct linguistic processing of the proposed character(s) or script. Examples of such properties are: Casing information, Numeric information, Currency information, Display behaviour information such as line breaks, widths etc., Combining behaviour, Spacing behaviour, Directional behaviour, Default Collation behaviour, relevance in Mark Up contexts, Compatibility equivalence and other Unicode normalization related information. See the Unicode standard at <http://www.unicode.org> for such information on other scripts. Also see Unicode Character Database (<http://www.unicode.org/reports/tr44/>) and associated Unicode Technical Reports for information needed for consideration by the Unicode Technical Committee for inclusion in the Unicode Standard.

¹ Form number: N4102-F (Original 1994-10-14; Revised 1995-01, 1995-04, 1996-04, 1996-08, 1999-03, 2001-05, 2001-09, 2003-11, 2005-01, 2005-09, 2005-10, 2007-03, 2008-05, 2009-11, 2011-03, 2012-01)

C. Technical - Justification

1. Has this proposal for addition of character(s) been submitted before?	Yes
If YES explain	<i>Replaces L2/10-207, L2/12-235, L2/15-005</i>
2. Has contact been made to members of the user community (for example: National Body, user groups of the script or characters, other experts, etc.)?	Yes
If YES, with whom?	<i>Mukund Gokhale (Script Research Institute, Pune) Motiravan Kangle (Akhil Gondwana Gondi Sahitya Parishad, Nagpur)</i>
If YES, available relevant documents:	
3. Information on the user community for the proposed characters (for example: size, demographics, information technology use, or publishing use) is included?	Yes
Reference:	
4. The context of use for the proposed characters (type of use; common or rare)	Common
Reference:	<i>Used for writing the Gondi language in India</i>
5. Are the proposed characters in current use by the user community?	Yes
If YES, where? Reference:	
6. After giving due considerations to the principles in the P&P document must the proposed characters be entirely in the BMP?	N/A
If YES, is a rationale provided?	
If YES, reference:	
7. Should the proposed characters be kept together in a contiguous range (rather than being scattered)?	Yes
8. Can any of the proposed characters be considered a presentation form of an existing character or character sequence?	No
If YES, is a rationale for its inclusion provided?	
If YES, reference:	
9. Can any of the proposed characters be encoded using a composed character sequence of either existing characters or other proposed characters?	No
If YES, is a rationale for its inclusion provided?	
If YES, reference:	
10. Can any of the proposed character(s) be considered to be similar (in appearance or function) to, or could be confused with, an existing character?	No
If YES, is a rationale for its inclusion provided?	
If YES, reference:	
11. Does the proposal include use of combining characters and/or use of composite sequences?	Yes
If YES, is a rationale for such use provided?	Yes
If YES, reference:	<i>Combining signs</i>
Is a list of composite sequences and their corresponding glyph images (graphic symbols) provided?	
If YES, reference:	
12. Does the proposal contain characters with any special properties such as control function or similar semantics?	Yes
If YES, describe in detail (include attachment if necessary)	<i>Virama;</i>
	<i>see text of the proposal</i>
13. Does the proposal contain any Ideographic compatibility characters?	No
If YES, are the equivalent corresponding unified ideographic characters identified?	
If YES, reference:	